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Robustness Assessment of Public Transport Networks in Various Graph Representations: Systematic Review, Decision Support, and Case Study

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ABSTRACT

The analysis of certain properties of the underlying graph of a public transport network generates insights about the network's structure. Hereby, the choice of the graph representation depends on a trade-off between complexity reduction and information preservation to adequately model a public transport network. The structure of the graph determines the robustness of the network and how resilient it is to disturbances like the breakdowns of stations. In this article, local and global properties of a graph are examined, which can be utilized as robustness indicators for long-term infrastructure planning, as well as the interplay with the choice of the graph representation, data availability, and assessment method. Furthermore, decision support is provided to guide the evaluation of the robustness of real-world public transport networks from multiple perspectives using open data sources. These concepts are illustrated in a case study for the rapid transit and subway system in Hamburg, Germany.

1 | Introduction

Public transport is increasingly relevant due to social, economic, and environmental factors [1]. With ongoing urbanization and population growth, cities become more densely populated, making efficient mobility solutions essential. Public transport infrastructure can enhance social equity [2, 3] by improving mobility for disadvantaged individuals [4, 5] and increasing job accessibility [6, 7]. Environmentally, public transport reduces carbon emissions, air pollution [8, 9], and traffic congestion [10, 11], contributing to more sustainable urban environments.

Given the importance of public transport, it is crucial to ensure that public transport networks (PTNs) remain resilient to disruptions such as cancellations, overcrowding, or accidents to

maintain reliable service. Reliability is a key factor in increasing public transport usage [12] and enhancing user satisfaction [13]. While operational strategies like vehicle and crew rescheduling can help mitigate disruptions [14], the structural design of a PTN impacts its robustness in the long term [15]. For instance, a subway system reliant on a single central line may experience a complete breakdown if no alternative routes exist. Identifying such bottlenecks is essential when planning new networks or expanding existing ones [16]. Since network design not only affects service reliability but also involves high costs and long implementation timelines, strategic planning is vital to ensuring sustainable and efficient public transport infrastructure [17].

Graphs have proven to be valuable tools for modeling and analyzing various real-world systems, including biological, social,

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financial, and transportation networks [18]. A PTN can be effectively represented as a graph, allowing for the analysis of its robustness, vulnerability, and resilience to disruptions [19]. However, these concepts are defined differently across PTNs, making it challenging to establish a unified characterization. For instance, the definition of robustness depends on the operator's priorities. One goal might be to minimize the number of disrupted routes, ensuring overall network stability. Alternatively, the focus could be on maintaining service for the highest number of passengers, prioritizing the most frequently used routes over general stability. Defining these objectives is essential before assessing robustness. This lack of consensus on a precise definition has led researchers to focus instead on quantitative indicators of these properties [20–22]. Rather than seeking a single definition, identifying meaningful metrics is more practical. General discussions on graph vulnerability and robustness can be found in [23], whereas [24] explores various robustness metrics for general networks.

Early applications of graph theory and network science to PTNs date back to the late 1950s, with a comprehensive summary available in [25]. The introduction of graph-based transit network indicators was first proposed by Garrison and Marble [26], including the α -index for determining the maximum number of cycles, the γ -index as a connectivity measure, and the β -index for assessing network complexity. Other early indicators incorporated graph properties into public transport analysis, such as network diameter, which approximates the shortest travel time between two terminal stations. By the 1980s, research began integrating trip-time-related properties into transit system models [27]. Later work, such as [28], categorized various graph indicators into state, form, and structure to describe metro PTNs. Studies on transport system vulnerability and resilience emerged as well, with [29] discussing general transport systems and [30] summarizing resilience metrics for railway networks based on research from 2008 to 2019. Additional reviews, such as [31], examine graph metrics for bus and metro PTNs up to 2019, while [32, 33] provide more recent surveys on urban rail transit resilience. A broader overview of traffic network analysis is available in [34].

In this article, we investigate how graph-based properties of a PTN can quantify its resilience to disturbances. Specifically, we explore which graph metrics serve as relevant indicators of robustness across different transportation modes. Our contribution lies in providing decision support to identify various strategies for assessing PTN robustness, including comparing indicators across different PTNs to demonstrate their use in quantifying system resilience for long-term infrastructure planning. While previous studies often relied on a single graph modeling approach, assessment method, or a limited set of indicators, we adopt a more holistic approach. The central question we address is: Given openly available data on a PTN, which graph representation, assessment method, and indicators can be used for evaluating its robustness based on a specific objective? Since there are various ways to model a PTN, it is important to consider the trade-off between reducing complexity and preserving information to ensure adequate analysis, which depends on the availability and quality of open data. By comparative analysis across different cities, a decision-support system for assessing the robustness of PTNs can become a valuable tool for city planners, enabling

the evaluation of efficient network designs globally and promoting improvements through insights drawn from other urban contexts. In addition, the impact of infrastructure changes, such as new line extensions, can be evaluated through a before and after assessment.

This article is structured as follows. In Section 2, concepts from graph theory are described which can be used for assessing robustness of PTNs. A systematic literature review of data sources needed for modeling real-world public transport systems with graphs, as well as robustness indicators is presented in Section 3, and how the graph representation and assessment method provide different insights. Based on these findings, a decision-support approach for modeling and assessing robustness of PTNs is provided in Section 4. In Section 5, these concepts are illustrated within the scope of a case study for the subway and rapid transit system in Hamburg, Germany. This article ends with a conclusion in Section 6.

2 | Modeling Public Transport Networks With Graphs

In this section, key definitions and concepts from graph theory and network science relevant to modeling PTNs as graphs are introduced. While some properties are traditionally associated with one field or the other, significant overlaps exist. Therefore, we do not differentiate their origins but refer to [35] and [36] for formal definitions.

2.1 | Graph-Theoretic Concepts

In the following, concepts from graph theory are introduced to establish uniform notation throughout the article and to provide the foundation for defining robustness metrics.

2.1.1 | Graph Representation

A PTN can be modeled by a graph $G = (V, E)$ with $V = \{v_1, \dots, v_n\}$ being the set of vertices or nodes (e.g., stations or stops), and $E = \{e_1, \dots, e_m\}$ being the set of edges or links (e.g., connections between stations) of G , where $e_k = \{v_i, v_j\}$, $v_i, v_j \in V$, and $e_k \in E$. Directed edges (arcs) can be used to describe one-way routes, where the ordered pair $e_k = (v_i, v_j)$ indicates that v_i is the predecessor of v_j .

2.1.2 | Node Degree

The degree $\deg(v_i)$ of a node v_i is the number of incident edges, whereas $\deg_k$ refers to the number of nodes with degree k . The neighborhood of a node $v_i \in V$ is defined as $N(v_i) = \{v_j \in V | \{v_i, v_j\} \in E\}$, indicating stations that are one stop apart. The k -core graph is a reduction of a graph where every node of degree $k - 1$ is successively removed. The largest connected component (LCC) (also called giant component) is the largest subgraph of a graph which is connected.

2.1.3 | Paths, Cycles, and Trees

A path $P = (v_i, \dots, v_j)$ is a sequence of distinct nodes where each successor node is adjacent to its predecessor, forming a connected subgraph of G . This path can be interpreted as a route or service line in a PTN. The distance $d(v_i, v_j)$ represents the shortest path between two nodes v_i and v_j . The symbol σ_{ij} denotes the number of shortest paths between nodes v_i and v_j , while $\sigma_{v_i v_k}(v_j)$ indicates the number of shortest paths between v_i and v_k that pass through v_j . A cycle is a path in which the first and last vertices are the same, whereas a triangle is a cycle that consists of exactly three nodes and three edges. If a connected graph contains no cycles, it is called a tree.

2.1.4 | Spectral Graph Theory

The entries of an adjacency matrix $A \in \{0, 1\}^{n \times n}$ for an undirected graph G with $|V| = n$ are defined as $a_{i,j} = 1$ if nodes v_i and v_j are adjacent, and $a_{i,j} = 0$ otherwise. In the case of a directed graph, $a_{i,j} = 1$ if there is an arc $(v_i, v_j) \in E$, which results in a non-symmetric adjacency matrix. A variation of the adjacency matrix is the Laplacian matrix $L = D - A$, where D is the diagonal degree matrix with $d_{ii} = \deg(v_i)$ for $i = j$, and $d_{ij} = 0$ for $i \neq j$. These matrices are of particular interest because the eigenvalues $\lambda_1, \dots, \lambda_n$ of the adjacency matrix, as well as the eigenvalues $\mu_1, \dots, \mu_n$ of the Laplacian matrix, provide insights into the structure of the graph.

2.1.5 | Weighted Graphs

Node weights $w(v)$ and edge weights $w(e)$ can provide additional information about a PTN. For example, $w(e)$ can represent the travel time between two stations or the average number of passengers traversing a particular connection, commonly referred to as passenger flow.

2.2 | Graph Properties

Certain graph properties serve as indicators of robustness and vulnerability, which can be classified into local and global properties. Local properties describe specific nodes or edges within the graph. Commonly referenced measures from the literature include various forms of degree, closeness, betweenness, PageRank centrality, and clustering coefficients, as outlined in Table 1. These metrics can similarly be applied to edges. On the other hand, global graph indicators, presented in Table 2, reflect overall characteristics of the entire graph, enabling comparisons between different graphs. In this article, these indicators are categorized into graph-based and spectral-based types, following a classification similar to that in [23].

2.2.1 | Graph-Based Indicators

Graph-based indicators include the average degree $\langle k \rangle$, the degree diversity κ (also known as Molloy-Reed parameter), and the degree distribution $P(k)$. A graph is considered scale-free if the degree distribution follows a power law, i.e., $P(k) \sim k^{-\gamma}$, with

TABLE 1 | Considered local graph properties for nodes and formula.

Property	Formula
Degree centrality	$C_D(v_i) = \frac{\deg(v_i)}{n-1}$
Closeness centrality	$C_C(v_i) = \frac{1}{\sum_{v_j \neq v_i} d(v_i, v_j)}$
Betweenness centrality	$C_B(v_i) = \sum_{v_j \neq v_i \neq v_k} \frac{\sigma_{v_j v_k}(v_i)}{\sigma_{v_j v_k}}$
PageRank centrality	$C_P(v_i) = \alpha \sum_j a_{ji} \frac{C_P(v_j)}{L(j)} + \frac{1-\alpha}{n}$ with $L(j) = \sum_i a_{ji}$ and $\alpha \in (0, 1)$
Clustering coefficient	$CC(v_i) = \frac{2 \{e_{jk} : v_j, v_k \in N(v_i), e_{jk} \in E\} }{ N(v_i) (N(v_i) -1)}$

Note: These properties can be defined similarly for edges.

the exponent γ being called the scaling factor. The degree assortativity ρ measures the tendency of nodes in a network to be connected to other nodes with similar degrees, whereas $\langle c \rangle$ is the average of all nodes' clustering coefficients. Other indicators involve path length characteristics, such as the average shortest path length $\langle l \rangle$ and the diameter $\langle d \rangle$.

2.2.2 | Spectral-Based Indicators

Spectral-based indicators refer to global graph properties that involve eigenvalues in some form. The natural connectivity $\langle n \rangle$ is derived from the eigenvalues of the adjacency matrix, while the algebraic connectivity $\langle \mu \rangle$ and effective graph conductance C_G are based on the eigenvalues of the Laplacian matrix. These are normalized versions to ensure comparability of values across PTNs of different sizes. In the following sections, it will be shown how these metrics can be used to assess the robustness of PTNs.

3 | Literature Review

In this section, approaches from the literature to evaluate the robustness of PTNs in different graph representations are reviewed. The choice of the topological representation, or in other words, in which space the PTN should be embedded, depends on the structure of the PTN, data availability, desired complexity and information preservation, as well as the objective of the robustness assessment. As different graph models can yield different insights, it is not trivial which representation should be selected. Usually, PTNs include a variety of transportation modes on rail and on road, like metro, subway or bus, and even a single transportation mode can include several variants, such as standard or express bus lines. The isolated view on a single transportation mode might not realistically model the interconnectedness of different transportation modes. Also the aggregated view of multiple transportation modes in a single graph might not be appropriate to capture the inherent characteristics of the modes. Therefore, it is important to recognize the trade-off between complexity reduction and information preservation. On the one hand, the reduction of the complexity might make it easier to identify the most important bottlenecks in a PTN. On the other hand, if the model is too simple, relevant information which is essential to capture the inherent characteristics of a PTN might not be included.

TABLE 2 | Considered global graph properties and formula with the corresponding category.

Category	Property	Formula
Graph-based	Average degree	$\langle k \rangle = \frac{2m}{n}$
	Degree diversity	$\kappa = \frac{\sum_{i=1}^n \deg(v_i)^2}{\sum_{i=1}^n \deg(v_i)}$
	Scaling factor	$P(k) \sim k^{-\gamma}$ with $P(k) = \frac{\deg_k}{n}$
	Degree assortativity	$\rho = \frac{\sum_{i,j} (\deg(v_i) - \langle k \rangle)(\deg(v_j) - \langle k \rangle)}{\sqrt{\sum_i (\deg(v_i) - \langle k \rangle)^2 \sum_j (\deg(v_j) - \langle k \rangle)^2}}$
	Average clustering coefficient	$\langle c \rangle = \frac{1}{n} \sum_{i=1}^n CC(v_i)$
	Average shortest path length	$\langle l \rangle = \frac{1}{n(n-1)} \sum_{i \neq j} d(v_i, v_j)$
	Diameter	$\langle d \rangle = \max_{i,j} d(v_i, v_j)$
Spectral-based	Natural connectivity	$\langle n \rangle = \frac{1}{n - \log(n)} \log(\frac{1}{n} \sum_{i=1}^n e^{\lambda_i})$
	Algebraic connectivity	$\langle \mu \rangle = \frac{\mu_{n-1}}{n}$
	Effective graph conductance	$C_G = \frac{n-1}{R_G}$ with $R_G = n \sum_{i=1}^{n-1} \frac{1}{\mu_i}$

First, various data sources that can be used to construct various graph models are discussed. Then, robustness assessment in $\mathbb{L}$ -space representation is reviewed. Next, approaches for multi-layer graphs in $\mathbb{M}$ -space representation are illustrated and last, the $\mathbb{P}$ -, $\mathbb{B}$ -, and $\mathbb{C}$ -space representations are considered.

In Figure 1, these concepts are illustrated. On the upper left corner, an exemplary PTN with two transportation modes (bus and subway) and three different routes, a bus route B (green), as well as two subway routes S1 (blue) and S2 (red) is considered. In (a) $\mathbb{L}$ -space, stations or stops are modeled as nodes $v_1, \dots, v_6$, and connections between consecutive stations are represented as edges. In the case of the multimodal exemplary PTN, nearby stations of different transportation modes are aggregated into a single node, and connections of different modes or of different routes of the same mode are aggregated into a single edge, leading to a reduced complexity. In contrast, the (b) $\mathbb{M}$ -space representation yields to a multilayer graph, where each transportation mode or route is represented as a separate layer. Interlayer edges connect nodes across layers to represent multimodal stations that are either shared by multiple modes or located in close proximity, preserving more information at the cost of increased complexity. In contrast to $\mathbb{L}$ - and $\mathbb{M}$ -space, graph representations in $\mathbb{P}$ -, $\mathbb{B}$ -, and $\mathbb{C}$ -space focus on entire routes rather than individual stations and connections. In (c) $\mathbb{P}$ -space, nodes represent stations and are connected if they are serviced by at least one common route in the PTN, forming complete subgraphs for each route. For clarity, only the induced subgraph of the bus route is highlighted in green. In (d) $\mathbb{B}$ -space, both stations and routes are represented as distinct node types in a bipartite graph. Station nodes are linked to route nodes if the corresponding stations are part of those routes in the PTN. The colored square nodes represent routes, while station nodes are linked to the routes they belong to. For example, since the green bus route consists of stations v_1, v_3, v_4 , and v_5 , the green square node is connected to these. The most aggregated and least complex modeling approach is found in (e) $\mathbb{C}$ -space. Here, only route nodes exist, and edges between them indicate whether a transfer is possible between the corresponding routes. In other words, route nodes are connected if at least one station in the PTN is served by both routes. The green, blue, and red square nodes

Public Transport Network

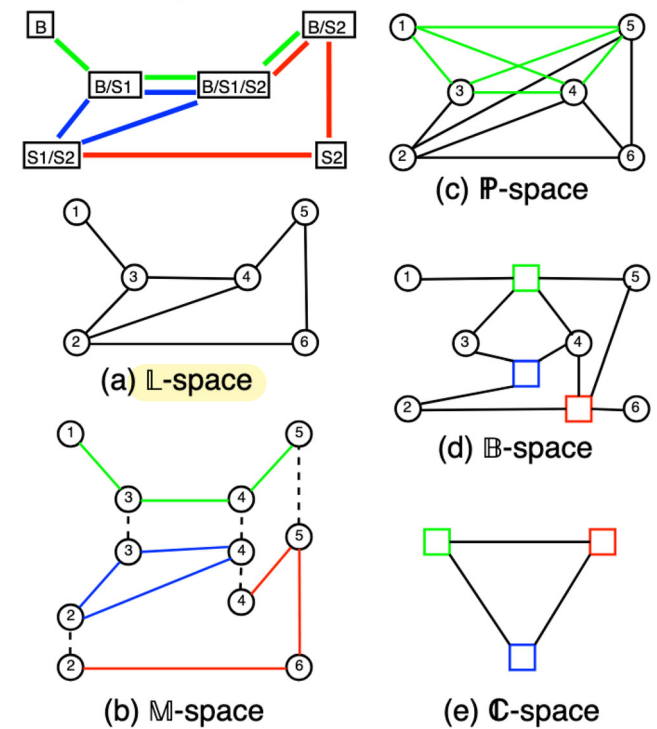


FIGURE 1 | Representation of a PTN with two transportation modes (bus and subway) and three different routes, a bus route B (green), as well as two subway routes S1 (blue) and S2 (red), in $\mathbb{L}$ -, $\mathbb{M}$ -, $\mathbb{P}$ -, $\mathbb{B}$ -, and $\mathbb{C}$ -space representation.

represent their respective routes and are interconnected because all three routes share at least one station.

Transformations between these representations follow a natural hierarchy of abstraction. The $\mathbb{M}$ -space representation is closest to the original PTN, as it preserves routes, modes, and multimodal station relationships in separate layers. From $\mathbb{M}$ -space, the corresponding $\mathbb{L}$ -space representation can be obtained by projecting all layers onto a single layer, merging nodes that represent the same

station across modes, and aggregating parallel edges that represent consecutive stations within any route. The $\mathbb{P}$ -space representation can be constructed from $\mathbb{L}$ -space by connecting all stations that share at least one route, forming complete subgraphs for each route. The $\mathbb{B}$ -space representation can be derived by identifying all station-to-route memberships and creating a bipartite graph linking each station to the routes it belongs to. Finally, $\mathbb{C}$ -space can be obtained by projecting the $\mathbb{B}$ -space representation onto the set of route nodes, connecting two route nodes whenever they share at least one adjacent station.

3.1 | Open Data Acquisition for Graph Modeling

In this section, approaches for collecting and transforming real-world data into PTN graph models are identified. Table 3 presents different types of data sources, highlighting their advantages and limitations, along with examples of studies that utilize them. A key focus is on making the most of openly accessible data, as many transportation operators are unwilling to share data for research purposes without complex agreements, if they provide access at all. To address these limitations, various graph representations can be leveraged to extract the highest value from available open data sources and to enable flexible modeling despite data constraints.

Previous research, such as [37], highlighted the limited availability of open-access PTN data, leading researchers to develop their own data collection methods. For instance, [38] partially relied on the book “Transit Maps of the World” [39] to model metro systems as graphs. Similarly, Yang et al. [40] obtained data from the official Beijing subway website, converting station and

route information into a WGS-84 coordinate system. The analysis in [41] is based on data from a Paris railway operator, while [42] utilizes data from the Land Transport Authority of Singapore, and [43] relies on network data from the City of Amsterdam. For smaller networks, relying on static maps may be sufficient to construct a simplified graph representation, as demonstrated in [44]. Particularly for bus PTNs, automatic vehicle location (AVL) systems, typically utilizing global positioning system (GPS) data, are employed to track vehicle trajectories and speeds, as demonstrated in, for example, [45–52]. With automatic passenger counting (APC) data, passenger flow can be integrated as edge weights or as components of an edge feature vector, typically represented in an origin-destination demand matrix. It is important to note that earlier studies, such as [53–55], obtained such data directly from the transportation authorities of the respective PTNs. This limits open data access and makes it difficult to reproduce the results of these studies. Similarly, Sun et al. [56] utilized automated fare collection (AFC) data to simulate passenger flow. In contrast, [57] accessed publicly available daily commute data for San Francisco from the U.S. Census Bureau which differentiates between network and operation data for a PTN. The operation data extends the information with timetable and APC data, which can be modeled with a directed, weighted graph, or even with multiple weights represented as node or edge vectors. For instance, in [55], APC data from passenger smart cards was used to construct weight matrices for both the rail and bus networks.

Approaches relying on data from transportation agencies are limited by the fact that such data is often not openly accessible or only available upon request, with no guarantee of approval, thereby hindering the reproducibility of results. Moreover, these methodologies are not universally applicable to other PTN systems due to the lack of a standardized data format. However,

TABLE 3 | Data sources which can be used to construct graphs from PTNs.

Data	Source	Advantages	Limitations	Studies
Public transport maps	Public transport agency website	Easy to access and interpret	Not suited for automated extraction	[38, 40, 44]
Data repositories	Various sources	Aggregated from multiple systems	Limited or inconsistent coverage	[58, 59]
Geographic information system (GIS) data	Google Maps, OpenStreetMap, Baidu Map, Amap	Offers detailed spatial information	May lack real-time or service-specific data	[57, 60–70]
Automatic vehicle location (AVL) GPS data	Public transport operators	Enables real-time tracking	Access depends on cooperation from operators	[45–52]
Automated passenger count (APC) data	Smart card data from public transport operators	Provides detailed passenger flow insights	Access restricted and reliant on operators	[47, 53–55, 66, 70]
Automated fare-collection (AFC) data	Smart card data from public transport operators	Useful for travel behavior analysis	Often inaccessible due to operator restrictions	[50, 51, 56, 69, 71–76]
General transit feed specification (GTFS) data	Public transport agency websites	Standardized and openly available	Often lacks ridership or real-time data	[77–84]

with today's improved accessibility to open data sources and the growing availability of computational resources, analyzing public transport data across the globe has become easier [85], even without relying on external partners such as transportation operators. Data from providers such as Google Maps¹ and OpenStreetMap² can be directly used to construct graphs based on geographic information system (GIS) data. This approach has been applied in studies such as [57, 61–63, 65, 67]. But GIS data is often inconvenient to extract due to its heterogeneous formats and lack of standardization. In contrast, the general transit feed specification (GTFS) can be utilized to extract information on PTN structures for graph representation.³ A significant advantage of GTFS is its widespread adoption, with numerous public transport agencies sharing this data in an open and accessible format [86]. In contrast to APC and AFC data, which are often proprietary, subject to privacy concerns, and rarely shared publicly, GTFS provides a standardized, openly available source that facilitates reproducibility and comparative research across different PTNs. For instance, [81] provides an overview of graph transformations using GTFS data, while the framework in [80] utilizes GTFS as the foundation for analyzing PTNs. Other examples of GTFS-based PTN analyses can be found in [77–79, 82, 83]. Due to data restrictions from transportation agencies, network properties can approximate passenger flow distribution in PTNs. This approach offers a suitable alternative to traditional passenger estimation and prediction models, which often depend on numerous assumptions and are constrained by limited data availability [87–89]. This strengthens the usability of GTFS data for constructing graph-based models, where network indicators can not only approximate passenger demand but also provide insights into PTNs robustness.

Once a specific graph representation is chosen, it is possible to further aggregate or extend information about the PTN. Since most traffic in a PTN occurs at the center of the network, where disturbances are likely to have the greatest impact, it can be useful to analyze the cores of the graph. For example, [44] simplify the graph by focusing on the 2-core, as the peripheral lines of the PTN have very little impact on the rest of the network. Similar graph reductions can be found in [90]. In summary, open GTFS data

is particularly useful for long-term infrastructure planning, as it provides standardized, regularly updated information on transit schedules and network structures without requiring direct collaboration with transportation operators. For graph modeling, analysis, visualization, and simulation, various software tools can be utilized. These include ArcGIS (e.g., [65, 66, 91]), BusMezzo (e.g., [92]), Gephi (e.g., [65]), and SUMO (e.g., [65]). Additionally, several libraries, such as NetworkX and PathPy in Python or iGraph in R (e.g., [83, 93, 94]), are commonly used for these tasks.

3.2 | Assessment of Robustness in $\mathbb{L}$ -Space

Since the representations across different graph models vary significantly, assessment methods and performance indicators are reviewed separately, with attention to their distinct structural characteristics. First, static and dynamic assessment methods in $\mathbb{L}$ -space are reviewed.

3.2.1 | Static Assessment by Graph Properties

The static approach to assessing PTN robustness involves analyzing global properties of the underlying graph. Comparing different PTNs using these metrics helps to identify which networks exhibit greater robustness. Furthermore, evaluating system robustness before and after construction activities, such as new line extensions, allows for assessing the impact of infrastructure investments. In this section, we review studies that have evaluated similar global properties, as summarized in Table 2. In Figure 2, two exemplary PTNs are compared, which can be classified as more and less robust. Selected global indicators are calculated to quantify the intuitive differences, namely that the former is more decentralized and mesh-grid like, while the latter exhibits a more centralized, tree-like structure. First, the robustness conclusions derived from the graph-based indicators described in Section 2.2.1 are presented.

3.2.1.1 | Average Degree. For the average degree $\langle k \rangle$, typical values are in the range [1.93, 10.00]. This metric gives already

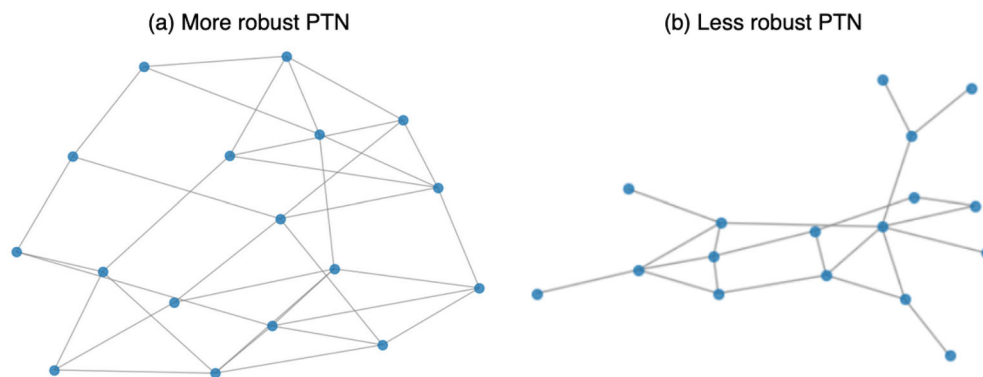


FIGURE 2 | A comparison between two exemplary PTNs, each consisting of 17 stations, illustrates how structural differences influence network robustness. In (a), the PTN exhibits a decentralized, mesh-like structure and could be considered comparatively more robust than the PTN in (b). Its representative global properties, $\langle k \rangle = 4.00$, $\kappa = 4.15$, $\langle c \rangle = 0.24$, and $\langle l \rangle = 2.04$, reflect a high level of connectivity. In (b), a comparatively less robust PTN is depicted, characterized by a more centralized, tree-like structure. Its global properties, $\langle k \rangle = 2.30$, $\kappa = 3.30$, $\langle c \rangle = 0.09$, and $\langle l \rangle = 2.60$, align with the intuitive impression of a less resilient network. The lower average degree $\langle k \rangle$, reduced degree diversity κ , and lower value of the clustering coefficient $\langle c \rangle$ indicate a less cohesive topology, while the higher average shortest path length $\langle l \rangle$ suggests a less efficient network in terms of overall reachability.

insights about the robustness, as a higher average degree is associated with higher robustness due to a more cohesive network with potentially more connections and alternative routes [95, 96]. This is illustrated, for example, in [97] where PTNs of London and Paris are compared. By assessing various robustness indicators, it is concluded that the Paris PTN is more robust than the London PTN. One indication of this increased robustness is the higher average degree of $\langle k \rangle = 3.73$ for the Paris PTN compared to $\langle k \rangle = 2.60$ for the London PTN. In rail PTNs, the average degree remains close to 2 because most stations are part of linear routes, connecting primarily to their preceding and succeeding stations. The higher reported values, such as those in [98] and [99], can be attributed to the fact that bus PTNs are being analyzed. Bus networks typically have a higher average degree due to their larger route overlap at stations and a more flexible infrastructure that is not constrained by a fixed rail system.

3.2.1.2 | Degree Diversity. Another indicator associated with robustness of PTNs is the degree diversity κ , also known as Molloy-Reed parameter in the literature. Reported values are in the range of [2.35, 7.91]. Also in this case, a higher value is an indicator for a more robust PTN, whereas $\kappa > 2$ is the threshold for the existence of a giant component in the network. It also serves as an approximate indicator of the stability of a PTN during disturbance simulations. The higher the value of κ , the larger the number of nodes that need to be removed to disintegrate a network [58, 96]. For instance, this is also illustrated in [97], where the authors conclude that the considered Paris PTN with $\kappa = 6.93$ is more resilient towards disturbances than the London PTN with $\kappa = 3.22$.

3.2.1.3 | Scaling Factor. In order to quantify the number of high-degree nodes in a graph, the degree distribution is a useful metric. When the distribution follows a power law, a network is considered as scale-free, indicating the existence of major hubs in the PTN. Identified values for the scaling factor γ in the power law are in the range of [2.10, 5.52]. A smaller value of γ indicates higher resilience [97], as a high value leads to fewer major hubs in the PTN. In the case of breakdowns of these hubs, the connectivity of the network is decaying faster than for a network with an increased number of hubs, or higher “meshedness” of the network [100]. As noted by Criado et al. [101], a complete graph exhibits minimal vulnerability because every node is connected to all others, ensuring that the network remains intact even if some nodes are removed. In contrast, a star-like network is highly vulnerable, as it relies on one single central hub to connect all others such that the removal of this key node would fragment the network. Referring again to the case study of [97], the concluded more robust Paris PTN exhibits a smaller value $\gamma = 2.62$, whereas for the London PTN, $\gamma = 4.48$ is determined.

3.2.1.4 | Degree Assortativity. Degree assortativity ρ refers to the tendency of nodes in a network to be connected to other nodes with similar degrees. In the literature, the following value range is identified: $[-0.02, 0.40]$. Assortativity has a direct relationship with the robustness of a PTN in terms of its connectivity, whereas higher values indicate more robust networks [102]. In an assortative PTN, hubs tend to be connected to other hubs. Therefore, if such high-degree nodes fail, the remaining high-degree nodes can still maintain connections with each other, reducing the chances of the network becoming disconnected [103].

3.2.1.5 | Average Clustering Coefficient. For the average clustering coefficient $\langle c \rangle$, a measure of how likely it is that two neighbors of a node are also connected to each other, the value range of $[0.00, 0.16]$ can be identified. A higher average clustering coefficient indicates a more cohesive network with more triangles, leading to a more robust PTN [23]. In particular, [96] find a strong correlation between the clustering coefficient and other robustness metrics. This can also be illustrated by an intuitive example; a tree graph has a clustering coefficient of 0 because it contains no triangles or cycles, whereas for a complete graph $\langle c \rangle = 1$.

3.2.1.6 | Average Shortest Path Length. For the average shortest path length $\langle l \rangle$, reported values are in the range of $[1.00, 53.50]$. Most of these studies agree that in terms of robustness a lower average shortest path length is preferable, as it can be an indicator for more available alternative routes. This is also observed for the Paris PTN with $\langle l \rangle = 6.40$ in comparison to the London PTN with $\langle l \rangle = 26.50$, further strengthening the argument that the Paris PTN is more robust [97]. Furthermore, the small-world property $\langle l \rangle \propto \log(|V|)$, together with a higher clustering coefficient (see, e.g., [104, 105]), can also be found in more robust PTNs.

3.2.1.7 | Diameter. For the diameter $\langle d \rangle$, the value range of $[14, 238]$ is observed in various studies. A smaller diameter typically signifies a more robust PTN, indicating a tightly coupled network [23], which is also observed for the more robust Paris PTN with $\langle d \rangle = 28$ in comparison to the London PTN with $\langle d \rangle = 107$. Next, the robustness conclusions derived from the spectral-based indicators, as described in Section 2.2.2, are presented.

3.2.1.8 | Natural Connectivity. The identified value range for the natural connectivity $\langle n \rangle$ is $[0.004, 0.22]$, while various studies conclude that a higher value indicates better robustness. This can be explained by the fact that the eigenvalues of the adjacency matrix are related to the number of closed walks in a graph (see, e.g., [127]), and thus serve as an approximate indicator of redundancy or the number of alternative routes [23, 96]. For example, the subway PTNs of Barcelona and Madrid are compared in [125], where Barcelona exhibits greater robustness according to various analyses. Among other metrics, the higher value $\langle n \rangle = 0.0077$ for Barcelona compared to $\langle n \rangle = 0.0044$ for Madrid further confirms its greater resilience.

3.2.1.9 | Algebraic Connectivity. The reported value range for the algebraic connectivity $\langle \mu \rangle$ is $[0.00001, 0.073]$, where the higher the value, the harder it is to disintegrate the network. If the second smallest eigenvalue of the Laplacian matrix $\mu_{n-1} = 0$, then the graph is disconnected and also represents a lower bound for the node and edge connectivity, that is, the minimum number of nodes and edges needed to disconnect a graph, respectively [23, 96]. Moreover, Bell et al. [128] conclude that μ_{n-1} can be related to the connectivity of any subgraph which is especially useful for examining specific parts of a PTN. Returning to the example of Barcelona and Madrid, the derived values for algebraic connectivity, $\langle \mu \rangle = 0.00006$ for Barcelona and $\langle \mu \rangle = 0.00001$ for Madrid, further confirm the greater robustness of Barcelona’s subway PTN [125].

3.2.1.10 | Effective Graph Conductance. For the effective graph conductance C_G identified values are in the range of [0.0009, 0.25]. A higher value indicates a more robust PTN, as this metric reflects both the number and length of paths between node pairs [23, 96]. This is also confirmed in [125], where for the more robust Barcelona PTN, $C_G = 0.0022$, and for the Madrid PTN, $C_G = 0.0009$ is calculated, consistently showing that higher values for spectral-based indicators can signal greater robustness.

In Table 4, the identified value ranges for the robustness indicators are presented, along with a specification of whether a higher or lower value typically implies greater robustness for a PTN.

3.2.1.11 | Aggregated Robustness Indicators. Besides single graph-based and spectral-based indicators, several attempts have been made to define aggregated robustness metrics. Some of these are reviewed in [32]. Notable examples include metrics based on the cyclomatic number as a proxy for alternative paths [38], with modifications proposed in [96] and [126], the resilience index score derived from passenger flows [42], and a comprehensive evaluation index that aggregates metrics such as maximum degree, clustering coefficient, and betweenness centrality [98]. These studies demonstrate substantial variation in how robustness is measured, depending on the considered transport modes, the availability of data, and the aspects regarded as most important. As a result, there is no clear consensus on a single metric that can capture all dimensions of robustness. This lack of agreement likely stems from the inherent difficulty in defining concepts such as robustness, vulnerability, and resilience, as well as the challenge of quantifying them within a single indicator.

In addition to aggregated metrics, the previously mentioned papers, along with many other studies, also rely on multiple indicators during analysis, which highlights the need to employ several complementary methods when assessing PTN robustness. Furthermore, correlations among metrics, overlaps in what they measure, and conflicts between different robustness objectives further complicate the selection of a single indicator [95]. Another important consideration is the interpretability of

robustness metrics, since more complex aggregated indices can be difficult to explain to stakeholders and to visualize effectively [23]. This highlights the need to develop a customizable robustness index tailored to the structural characteristics of a given PTN, aligning with the goals of the robustness assessment, ensuring explainability, and accounting for the considered transport modes and available data.

3.2.2 | Dynamic Assessment by Simulations

In this section, dynamic approaches based on simulations for evaluating robustness are discussed. During a simulated attack, disruption, or disturbance scenario, removing nodes or edges with high values for the metrics listed in Table 1 is more likely to cause connectivity loss of a PTN than removing random nodes. To illustrate how the removal of nodes with high degree, closeness, betweenness, or PageRank affects different types of PTNs, these centrality values are computed for all nodes in the two exemplary PTNs from Figure 2 and shown in Figure 3. Darker tones represent lower centrality values, while brighter colors indicate higher ones. The red node with a black outline marks the highest value for that specific metric. In the comparatively more robust PTN, these highest values are distributed across several nodes, whereas in the less robust PTN the same node attains the highest value for all metrics. This observation reinforces the previous conclusions drawn from the global properties, such that the PTN in (a) can be regarded as more robust because multiple nodes share critical importance, while for the PTN in (b) a single, clearly identifiable point of failure exists. Random failures may result from technical malfunctions in the rail or vehicle, or from passenger and operator misbehavior. In contrast, targeted attacks or disruptions often involve deliberate infrastructure damage [40]. Other scenarios include extreme weather events, such as coastal flooding [129]. Since such attacks can affect not just individual stations but entire PTN segments, stress test simulations can also assess the impact of path or subgraph failures. Similarly, cascading effects from a single failed station may lead to widespread disruptions, as operators might halt traffic on an affected line until repairs are completed [22].

TABLE 4 | Identified values for graph indicators in $\mathbb{L}$ -space, specifying whether lower or higher values typically imply greater robustness.

Category	Robustness indicator	Identified value range	Robustness tendency	Studies
Graph-based	Average degree $\langle k \rangle$	[1.93, 10.00]	Higher	[37, 58, 83, 97–99, 101, 102, 106–109]
	Degree diversity κ	[2.35, 7.91]	Higher	[38, 58, 83, 97, 102]
	Scaling factor γ	[2.10, 5.52]	Lower	[37, 38, 40, 58, 97, 98, 106, 109–113]
	Degree assortativity ρ	[−0.02, 0.40]	Higher	[37, 58, 62, 102, 109, 110, 114]
	Average clustering coefficient $\langle c \rangle$	[0.00, 0.16]	Higher	[58, 96, 98, 108–110, 115–120]
	Average shortest path length $\langle l \rangle$	[1.00, 53.50]	Lower	[37, 58, 83, 97, 99, 101, 107–110, 114, 116–118, 120–124]
	Diameter $\langle d \rangle$	[14, 238]	Lower	[37, 58, 83, 97, 101, 107, 108, 114, 115, 122, 123]
Spectral-based	Natural connectivity $\langle n \rangle$	[0.004, 0.22]	Higher	[96, 102, 125]
	Algebraic connectivity $\langle \mu \rangle$	[0.00001, 0.073]	Higher	[96, 102, 125]
	Effective graph conductance C_G	[0.0009, 0.25]	Higher	[96, 102, 125, 126]

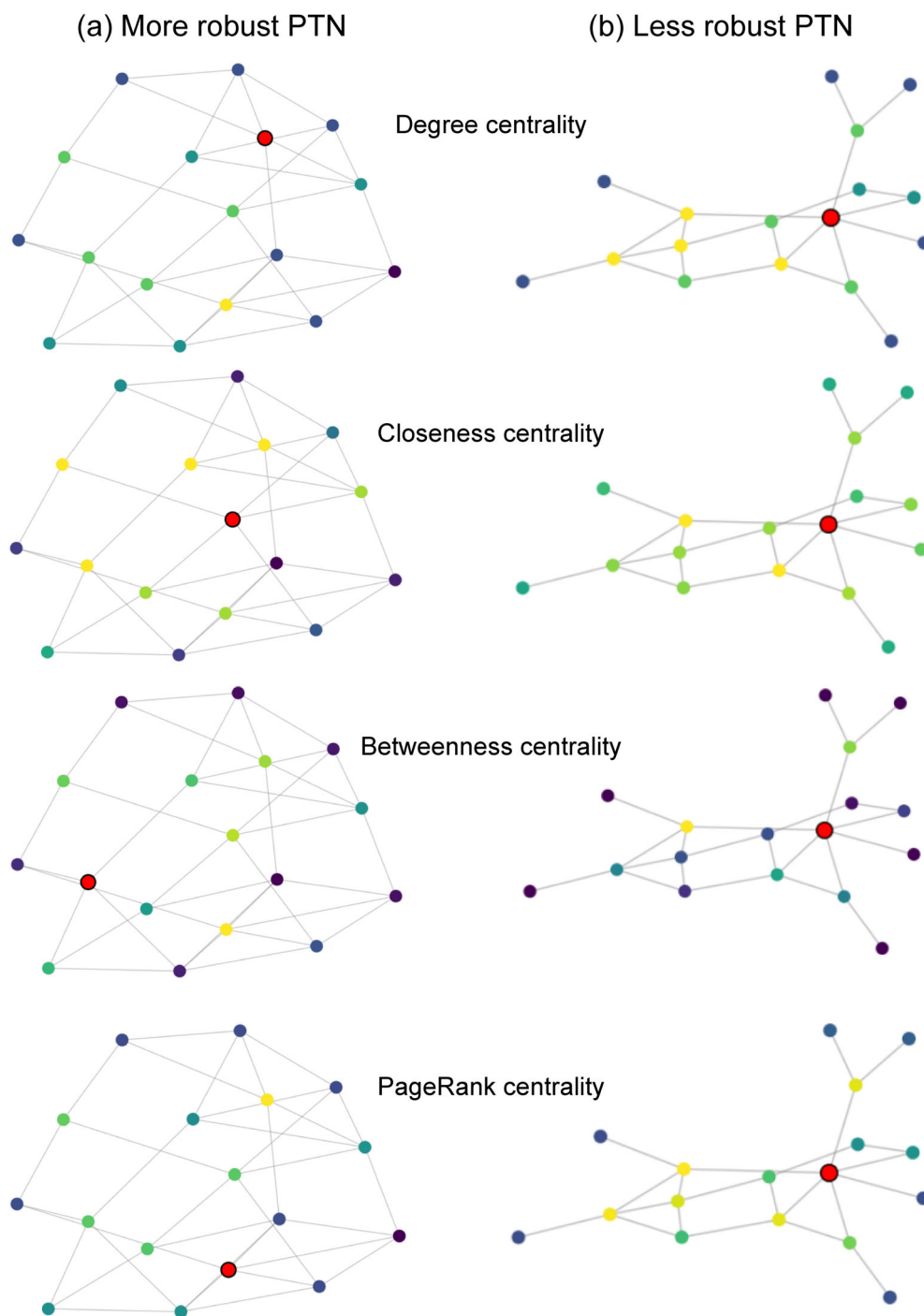


FIGURE 3 | Values for degree, closeness, betweenness, and PageRank centrality are shown for all nodes in the same exemplary PTNs as in Figure 2. Darker tones represent lower centrality values, while brighter colors indicate higher ones. The red node with a black outline marks the highest value for that specific metric. In (a), these maxima are distributed across several nodes in the more robust PTN, whereas in (b) the same node simultaneously attains the highest values for all metrics, highlighting a single point of failure.

The impact of sequentially removing vulnerable graph elements in simulations is typically assessed by how quickly key global graph properties decline. Local graph properties, such as node centralities, help identify vulnerable PTN components. For example, an attack targeting nodes with the highest degree will eventually lead to network disconnection. The size of the LCC in a fragmented graph serves as a resilience measure, representing the remaining operational portion of the PTN. A rapid decrease

in this size over multiple simulation steps indicates high vulnerability to the removal of high-degree nodes. An alternative approach tracks all connected components, monitoring the number of nodes after each attack iteration to provide a comprehensive view of the remaining functional PTN segments [61]. Another useful robustness metric is the average shortest path length [101], which reflects network efficiency during disruptions. As critical nodes or edges are removed, the shortest paths

between remaining node pairs tend to increase, indicating longer travel times within the PTN.

3.2.2.1 | Degree Centrality. Wang et al. [96] observed that, in some metro networks, complete network disintegration occurred nearly twice as fast when removing high-degree nodes compared to random removals. Similarly, Zhang et al. [113] confirmed the severity of such attacks by analyzing their impact on the average shortest path length. Tran et al. [112] found that networks were significantly more vulnerable in attack simulations targeting high-degree nodes, as multiple robustness metrics declined faster than in random failure scenarios. Xiao et al. [98] further concluded that node failures based on a high degree had the greatest impact on network stability, whereas removing nodes with high clustering coefficients had a much less severe effect. The relationship between node degree and natural connectivity was highlighted by Zeng et al. [68], who found that removing high-degree nodes led to a steep decline in relative natural connectivity. von Ferber et al. [97] also demonstrated that attacks on recalculated high-degree nodes had a more severe impact on PTN connectivity than random node removals. Other studies reinforce these findings. Golshanabadi and Popa [126] showed that targeted attacks on high-degree nodes were more disruptive than random attacks. In bus PTNs, [130] discovered that targeting high-degree nodes had a more severe impact on operating distance, service frequency, and passenger flow than attacks based on betweenness centrality.

3.2.2.2 | Closeness Centrality. Zhang and Ng [70] found that attacks based on closeness centrality had a relatively mild effect compared to strategies targeting degree, flow, or betweenness centrality. Zeng et al. [68] confirmed that removing nodes with high closeness centrality resulted in only a limited impact on the largest remaining component size, average shortest path length, and natural connectivity, whereas attacks targeting degree or betweenness centrality were far more disruptive.

3.2.2.3 | Betweenness Centrality. Frutos Bernal and del Rey [102] found that degree-based attack strategies caused damage comparable to attacks on nodes with high betweenness centrality. Similarly, Cats and Krishnakumari [122] observed that removing nodes and edges with high betweenness centrality caused the most disruption in three analyzed PTNs, followed by the removal of high-degree nodes and edges. Wu et al. [124] also found that betweenness-based attacks were more severe than random failures, leading to rapid PTN deterioration. This is also observed by Zhu et al. [131], concluding that nodes with a high betweenness centrality usually represent transfer stations which are vulnerable parts of a PTN. Wandelt et al. [132] analyzed attacks on nodes with the highest degree and betweenness centrality, with and without recalculating centrality after each iteration. They found that recalculating betweenness centrality had the most severe effect on robustness, though it was computationally expensive. Zhang et al. [133] further confirmed that nodes with the highest degree and betweenness centrality are the most vulnerable components of PTNs. In [40], a weighted combination of node degree and betweenness centrality was used to define node importance, showing that attacks on high-ranked nodes significantly reduced the LCC size in the remaining PTN. This suggests that combining multiple metrics may be more effective in identifying vulnerable stations than relying on a single indicator.

3.2.2.4 | PageRank Centrality. Zhang et al. [134] conclude that targeted attacks based on nodes with the highest PageRank are more severe than on nodes with the highest degree or betweenness centrality. This is also observed by Mouronte-López [83] for some of the analyzed PTNs.

3.2.2.5 | Route Disturbances. Zhang et al. [22] analyze whole-route failure scenarios rather than single-station or line disruptions. They assess route importance using the sum of node degree, node betweenness centrality, and edge betweenness. By aggregating these values across all stations or line segments within a route, they derive a single importance score. Their findings show that routes with the highest sum of node and edge betweenness centrality have a greater impact on network robustness than those selected randomly or based on node degree alone. Meng [135] applies a multi-criteria decision-making method to calculate node importance using degree, betweenness, closeness, and PageRank centrality. This approach is applied to scenarios involving disruptions at the node, edge, line, or area level. Similarly, Zhang et al. [136] distinguish between edge-based attack strategies that disrupt either a single route service (single-edge failure) or all connections between two stations (infrastructure failure).

3.2.2.6 | Passenger Flow. Abdelaty et al. [99] confirm that removing high-degree or high-betweenness centrality nodes has the most severe impact on network stability. However, they argue that such scenarios are unrealistic, as the simultaneous failure of major hubs is unlikely in a PTN. Instead, they analyze the impact of removing links with the highest flow, highlighting that traffic congestion is a more probable disruption in road networks. Fan et al. [137] emphasize the importance of incorporating dynamic passenger flow over time to assess station vulnerability more realistically. They note that disruptions have varying impacts depending on the time of day, station failures are more critical during rush hours than during off-peak times. Nevertheless, scenarios targeting high-degree stations still result in the greatest reduction in the size of the largest remaining component. Malandri et al. [138] focus on edges with high passenger volumes as attack targets, using average travel time delay (based on edge capacity) as a vulnerability indicator. Similarly, Chopra et al. [139] confirm that disruptions affecting high-passenger-flow routes are more severe than random failures. Xu et al. [140] find that passenger-flow-based attack strategies provide different insights compared to purely topological analyses. However, Sun et al. [111] argue that topological metrics, such as node degree and betweenness centrality, are better indicators of station vulnerability than passenger-flow-based metrics.

These findings suggest that targeted attacks on high-degree and high-betweenness centrality nodes often have the most severe impact on PTN stability, accelerating network fragmentation. While these topological metrics provide strong indicators of vulnerability, some studies argue that real-world disruptions are more likely to stem from high-passenger-flow bottlenecks rather than simultaneous hub failures. Incorporating dynamic passenger flow, traffic congestion, and travel time delays can offer a more realistic assessment of network resilience. Additionally, whole-route and edge-based disruptions further highlight the complexity of PTN vulnerability. Combining multiple metrics, rather than relying solely on topological indicators, may provide

the most comprehensive approach to identifying critical weaknesses and enhancing PTN robustness against various types of failures.

3.3 | Assessment of Robustness in M-Space

The complexity of a PTN can be captured using a multilayer graph, known as $\mathbb{M}$ -space, where each transportation mode or route is represented as a separate layer. Interlayer edges connect nodes across layers to represent multimodal stations that are either shared by multiple modes or located in close proximity. Figure 4 illustrates disruption scenarios for the simplified PTN shown in Figure 1. In the $\mathbb{L}$ -space representation, the impact of disruptions tends to be overestimated. For example, if subway station v_4 is removed due to a disturbance, the $\mathbb{L}$ -space model would suggest that the entire bus route is also affected. However, in $\mathbb{M}$ -space representation, only the affected subway layers (blue and red) are impacted, while the bus layer (green) remains operational. This distinction better reflects real-world scenarios where subway disruptions often lead to increased bus usage or the implementation of replacement bus services through inter-agency cooperation (usually called bus bridging; see, for example, [141, 142]). While the $\mathbb{M}$ -space approach might provide a more realistic representation of multimodal interactions for a dynamic robustness assessment, it introduces additional complexity. Some metrics applicable to simple graphs may no longer be directly usable, requiring adapted or new analytical approaches. Therefore, calculating the introduced indicators as part of a static robustness assessment is not feasible for PTNs in $\mathbb{M}$ -space. For instance, the average degree in $\mathbb{M}$ -space is not directly comparable to the average degree in $\mathbb{L}$ -space, as nodes can be connected by interlayer edges across different layers. These transfer edges must be interpreted differently from intralayer edges, which connect different stations within the same mode of transport. Another example is the calculation of eigenvalues, as the adjacency matrix and Laplacian matrix become the supra-adjacency matrix and supra-Laplacian matrix, respectively, requiring distinct analysis techniques tailored to the multilayer structure (see, e.g., [143] for a comprehensive overview).

3.3.1 | Complexity Reduction and Information Preservation

It is important to note that the layers in a multilayer graph do not need to follow the same structural representation. For instance, in [51], the metro network is modeled in $\mathbb{L}$ -space, while the bus network is represented in $\mathbb{P}$ -space. The choice of representation depends on the PTN's structure and the analysis objective, balancing complexity reduction with information preservation. This trade-off is discussed in [144], where aggregating multiple transportation modes into a simple graph is criticized for overlooking coupling effects and transfer times between modes. Similarly, Zhang et al. [145] examine how layer aggregation in a multilayer graph can maintain both topological structure and additional PTN attributes. Extending this approach, [75] introduce a fixed infrastructure layer and a time-dependent service layer to better capture dynamic passenger flows.

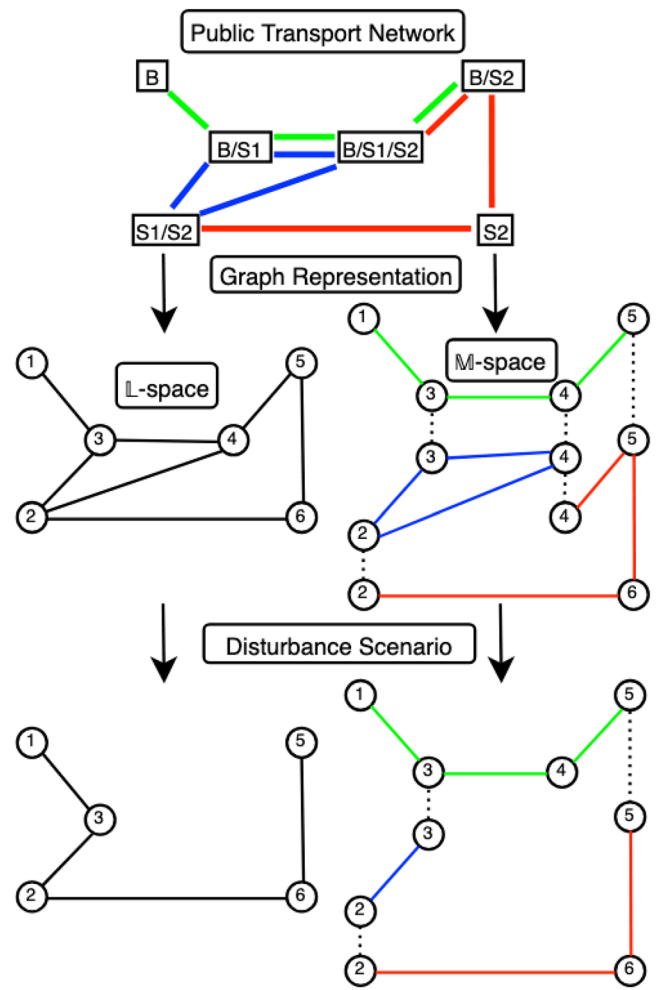


FIGURE 4 | Representation of a PTN with two transportation modes (bus and subway) and three different routes, B (green), S1 (blue) and S2 (red), in $\mathbb{L}$ - and $\mathbb{M}$ -space. The breakdown of the station v_4 of the S1 and S2 route in a disturbance scenario gives different insights in these spaces.

3.3.2 | Differences Compared to L-Space

The following discussion explores the implications of robustness assessments in $\mathbb{M}$ -space versus $\mathbb{L}$ -space representations of PTNs, further highlighting possible advantages of this approach. Xu et al. [91] and Zhang et al. [69] demonstrate that failing to account for interdependencies between different transportation modes can lead to inaccurate robustness assessments. Their findings, based on an interdependent bus-metro network with transfer possibilities, emphasize the importance of considering multimodal interactions. Similarly, Chen et al. [146] model a bus-metro PTN with transfer options and highlight the need for additional bus services during disruptions to improve network resilience. Chen et al. [147] conclude that multimodal PTNs are inherently more resilient to disturbances than single-mode networks, underscoring the importance of analyzing them as integrated systems. Li et al. [148] focus on the robustness of multilayer on-demand bus PTNs with uncertain topologies, demonstrating that the $\mathbb{M}$ -space representation effectively captures intermodal relationships. In [149], a rail-aviation PTN is analyzed, revealing that while the standalone rail network is less robust than the aviation network, viewing them as an integrated system provides a

more accurate robustness assessment. Likewise, [50] study a PTN comprising shared bikes, buses, and metro services, identifying metro stations as the most vulnerable components, an insight that would be difficult to obtain from analyzing the metro network in isolation.

3.3.3 | Similarities Compared to $\mathbb{L}$ -Space

In addition to the potential advantages of a $\mathbb{M}$ -space representation, several findings from the literature align closely with studies that utilize the $\mathbb{L}$ -space representation. Li et al. [150] find for an integrated metro, bus, taxi and shared bike PTN that high degree nodes are particularly vulnerable parts of the network. Pu et al. [151] also observe the scale-free characteristic of the considered PTN, whereas [72] confirm that a higher graph conductance indicates robustness. In [20] and [152], a highest degree-based attack strategy is more severe than random attacks, while [153] confirm that removal of nodes with a high degree and betweenness centrality causes the most impact during attack simulations. Lee et al. [72] find that nodes with a high recalculated betweenness and closeness centrality lead to the fastest robustness decline. Aparicio et al. [77] conclude that attack strategies based on the recalculated degree and betweenness centrality lead to the fastest decay of the considered multimodal PTN, similar to PTNs in $\mathbb{L}$ -space.

3.3.4 | Multilayer Structure

Liu et al. [73] examine various attack strategies that account for the multilayer structure of PTNs. Depending on the scenario, either a subway or bus station may fail or experience delays, which impacts passenger flows in distinct ways. Lee et al. [72] emphasize the significance of intermodal interdependence, arguing that the benefits of a more robust PTN outweigh the risks of cascading failures across multiple layers. In contrast, [152] observe that while passenger flow can be redirected to other modes during disturbances in interconnected PTNs, coordination challenges between different public transport operators sharing

the same infrastructure may undermine overall network stability. Baggag et al. [78] discuss how, in their bus and metro PTNs, the interdependence of nodes can cause cascading failures across layers, an effect that would be missed if each transportation mode were modeled separately. Similarly, Jin et al. [154] argue that aggregated models fail to capture key characteristics of passenger flow, such as transfers between different modes, which are critical to understanding the dynamics of PTN resilience.

3.3.5 | Cascading Effects

Ma et al. [20] demonstrate that failures in one layer can trigger cascading congestion effects throughout the entire network. When a failure occurs, passenger flow is redistributed not only to other stations within the same layer but also to stations in other layers. This can result in a station in the second layer reaching full capacity, potentially causing its failure as well. In contrast, [71] propose a different use of multilayer graphs, where one layer represents the underlying network structure and the other layer captures temporal passenger flow data, typically incorporated as edge weights. This approach allows for the simulation of different types of attack strategies: one targeting the infrastructure, such as station failures, and another focusing on passenger flow, as seen in congestion scenarios.

In Table 5, the main advantages and disadvantages of representations in $\mathbb{L}$ - and $\mathbb{M}$ -space are depicted, highlighting the trade-off between complexity reduction and information preservation.

3.4 | Assessment of Robustness in $\mathbb{P}$ -, $\mathbb{B}$ -, and $\mathbb{C}$ -Space

In contrast to $\mathbb{L}$ - and $\mathbb{M}$ -space, graph representations in $\mathbb{P}$ -, $\mathbb{B}$ -, and $\mathbb{C}$ -space focus on entire routes rather than individual stations and connections. This perspective provides valuable insights into transfer possibilities between routes. The breakdown of transfer stations can be critical, as limited transfer options may result in increased travel times within a PTN. Furthermore, for simulating

TABLE 5 | Comparison of $\mathbb{L}$ -space and $\mathbb{M}$ -space representations for PTNs.

Criteria	$\mathbb{L}$ -space	$\mathbb{M}$ -space
Information preservation	Aggregation of various transportation modes and routes can lead to an overly simplified representation of a PTN	Capturing multimodal interactions and transfer dependencies helps provide a more realistic model of a PTN
Analytical complexity	The simpler structure enables the application of graph metrics and established theoretical results for simple graphs, building on a well-researched foundation	The more complex structure requires adapted analytical techniques, such as supra-adjacency and supra-Laplacian matrices, to properly account for the multilayer nature of the network
Static assessment	Previous studies have calculated various robustness indicators for this representation, enabling extensive comparisons between new PTNs and existing values	Due to adapted definitions of various graph metrics for multilayer graphs, established robustness indicators for simple graphs can not be directly applied or compared
Dynamic assessment	Disturbances might be misinterpreted by assuming all modes or routes are affected simultaneously, potentially leading to an overestimation of disruption impacts	Independent disturbance scenarios within each mode or route can be simulated, enabling a more realistic modeling of failure propagation across different modes

TABLE 6 | Identified values for graph indicators in $\mathbb{P}$ - and $\mathbb{C}$ -space.

Space	Robustness indicator	Identified value range	Studies
$\mathbb{P}$	Average degree $\langle k \rangle$	[25.65, 236.65]	[37, 55, 83, 106, 108, 109, 116, 117, 158–161]
	Degree diversity κ	[73.20, 415.46]	[106, 160]
	Scaling factor γ	[2.91, 5.85]	[106, 160]
	Degree assortativity ρ	[−0.078, 0.06]	[55, 109, 116]
	Average clustering coefficient $\langle c \rangle$	[0.68, 0.95]	[108, 109, 116, 117, 158, 160, 161]
	Average shortest path length $\langle l \rangle$	[1.00, 4.79]	[37, 55, 83, 108, 109, 116, 117, 158–161]
	Diameter $\langle d \rangle$	[2, 14]	[37, 55, 83, 108, 159, 160]
$\mathbb{C}$	Average degree $\langle k \rangle$	[10.47, 151.72]	[37, 108, 116, 162, 163]
	Average clustering coefficient $\langle c \rangle$	0.66	[108]
	Average shortest path length $\langle l \rangle$	[1.56, 4.00]	[37, 108, 163]
	Diameter $\langle d \rangle$	[3, 10]	[37, 108]

Note: No values are provided for the $\mathbb{B}$ -space representation, as it lacks a clear interpretation in this context.

disturbances affecting entire routes, these representations could offer a more efficient modeling approach.

3.4.1 | $\mathbb{P}$ -Space Representation

The literature presents various motivations for modeling a PTN using $\mathbb{P}$ -space. From a passenger's perspective, the degree of a node in $\mathbb{P}$ -space represents the number of stations reachable without transferring to another route, while the shortest path between two stations indicates the number of required transfers. Sienkiewicz and Holyst [109] argue that analyzing routes provides deeper insights into the structural characteristics of a PTN than focusing solely on individual stations. Beyond robustness investigations in $\mathbb{L}$ -space, [106] demonstrate that the same analysis in $\mathbb{P}$ -space effectively captures transfer relationships between different routes, a finding also supported by [83, 117, 155]. Similarly, Soh et al. [55] model Singapore's rail and bus PTN, emphasizing that $\mathbb{P}$ -space is better suited than $\mathbb{L}$ -space for analyzing passenger movement. Cao et al. [156] utilize the $\mathbb{P}$ -space representation to focus on the robustness analysis of connections between city pairs. Similarly, Kopsidas and Kepaptsoglou [63] apply this representation to a substitution network of a metro PTN, indicating the number of alternative routes available to other stations. Feng et al. [157] further extend this approach by proposing a directed $\mathbb{P}$ -space graph.

The identified robustness indicator values are summarized in Table 6. Indicators such as the average degree, degree diversity, and clustering coefficient are higher in $\mathbb{P}$ -space compared to $\mathbb{L}$ -space, while the average shortest path length and diameter are lower. This confirms that PTNs modeled in $\mathbb{P}$ -space exhibit small-world properties [109]. For the average degree in $\mathbb{P}$ -space, which represents the number of reachable stations without transfers, high values are particularly observed in bus PTNs. For example, the bus networks of São Paulo with $\langle k \rangle = 137.46$ and Taipei with $\langle k \rangle = 236.65$ exhibit high mean route lengths (58.30 and 70.50, respectively). This, combined with greater route overlap at bus stations, explains their increased values [37], as well as for the degree-dependent values of the degree diversity κ [106]. In the event of disruptions affecting entire routes,

the number of reachable stations without transfers decreases, making the $\mathbb{P}$ -space representation a valuable model for evaluating robustness from a passenger's perspective.

This is illustrated in Figure 5, for the same exemplary PTN as in Figure 1, where a disruption of the bus route reduces the number of reachable stations without transfers at station v_3 , decreasing its degree from $\deg(v_3) = 4$ to $\deg(v_3) = 2$. In addition, the average shortest path length $\langle l \rangle$ would increase during route disturbance, signaling an increased average number of transfers for passengers. For instance, the shortest path between station v_3 and station v_5 has increased to $d(v_3, v_5) = 2$, whereas the stations could be reached without any transfers before.

3.4.2 | $\mathbb{B}$ -Space Representation

In $\mathbb{B}$ -space, [97] measure the impact of simulated cascading failures, where the breakdown of a station within a route affects the entire route. This is modeled by removing the corresponding route-type node. Likewise, [57] employ a bipartite usage graph to identify vulnerable segments within the PTN. Another bipartite graph-modeling approach is presented by [164], where metro stations form one node type and bus lines the other. This framework considers the bus network as a substitution service during metro system disruptions (bus bridging). Note that no values for the robustness indicators in $\mathbb{B}$ -space are provided in Table 6. One reason that these indicators are not calculated in the literature is that such metrics lack a clear meaning in this representation (see, e.g., [108, 165]). For example, the average node degree is not applicable due to the presence of different types of nodes. In Figure 5, the same bus route (shown as green, squared route type node) disruption scenario is illustrated in $\mathbb{B}$ -space. Unlike the $\mathbb{P}$ -space representation, the $\mathbb{B}$ -space provides a more coarse-grained view of the PTN, emphasizing which stations are still served by the remaining route type nodes. Furthermore, high-degree station type nodes are essential transfer stations that link multiple routes, making them particularly vulnerable stations in robustness assessments. For instance, in Figure 5, node v_4 acts as the sole bridge for all three routes, highlighting its role as a critical transfer station within the PTN. Therefore, an

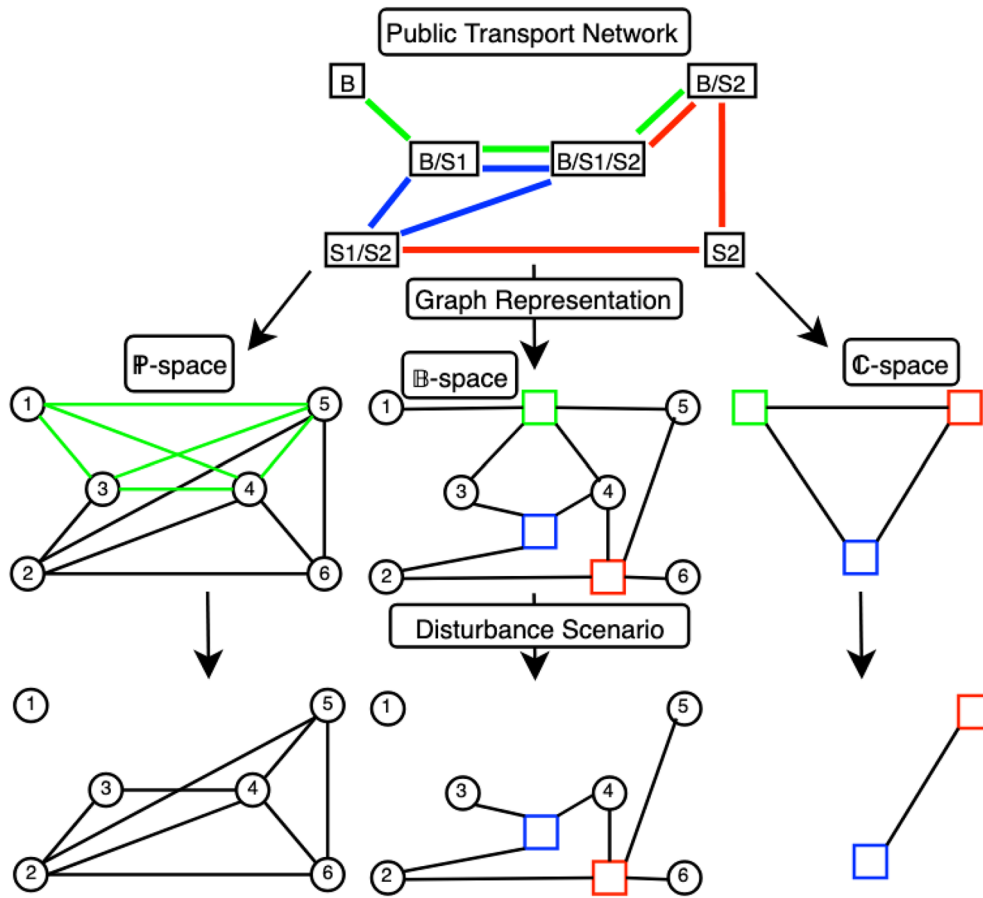


FIGURE 5 | Representation of a PTN with two transportation modes (bus and subway) and three different routes, B (green), S1 (blue) and S2 (red), in $\mathbb{P}$ -, $\mathbb{B}$ - and $\mathbb{C}$ -space. The different representations provide insights into the routes at varying levels of granularity. For clarity, only the induced subgraph of the bus route is highlighted in green for the $\mathbb{P}$ -space representation. Furthermore, the impact of disruptions to the green route is illustrated across the different spaces.

advantage of this representation is that it allows the robustness of a PTN to be assessed from two different perspectives simultaneously, focusing on both critical routes and stations. This benefit is also emphasized by [89, 108, 165].

3.4.3 | $\mathbb{C}$ -Space Representation

A more comprehensive, bird's-eye-view approach for analyzing PTN routes is to use the $\mathbb{C}$ -space representation. In this model, the shortest path between two nodes reflects the number of transfers required to travel from one route to another, while the degree of a node indicates the number of routes a passenger can transfer to [37]. Traveling within a given route of a PTN incurs no additional information cost for passengers, whereas transferring between lines requires extra information. Consequently, this representation reflects the information processing required to navigate within a PTN [116]. Sun et al. [162] argue that this representation is particularly useful for identifying critical links that act as structural bridges within the original PTN. Wang et al. [163] apply this representation to model a bus-line PTN, and conclude that targeted attacks are more disruptive than random failures, an observation also made in other representations. Similarly, Cicchini et al. [61] use this graph representation to model the failure of entire bus routes, emphasizing that, unlike

isolated node or edge failures, disturbances tend to propagate along an entire route, reflecting a more system-wide vulnerability assessment. Identified values for this representation are summarized in Table 6. Similar to $\mathbb{P}$ -space, due to the focus on routes, higher average degrees and lower values for both the average shortest path length and the diameter can be observed. In Figure 5, the previously illustrated bus route disruption scenario provides the simplest view of the impact in $\mathbb{C}$ -space, showing that transfers are still possible between the two remaining routes.

4 | Decision Support for Robustness Assessment of Public Transport Networks

In this section, the identified approaches for assessing the robustness of a PTN for infrastructure planning in a holistic manner are incorporated within a structured decision tree, as illustrated in Figure 6. This decision-support approach focuses on making the most of openly accessible and standardized GTFS data, as well as leveraging the usefulness of graph-based indicators, which have been shown to approximate passenger flows [87–89]. In doing so, it addresses the challenge of data restrictions imposed by transportation agencies. Furthermore, this approach distinguishes itself from existing methods such as [80], which rely on a

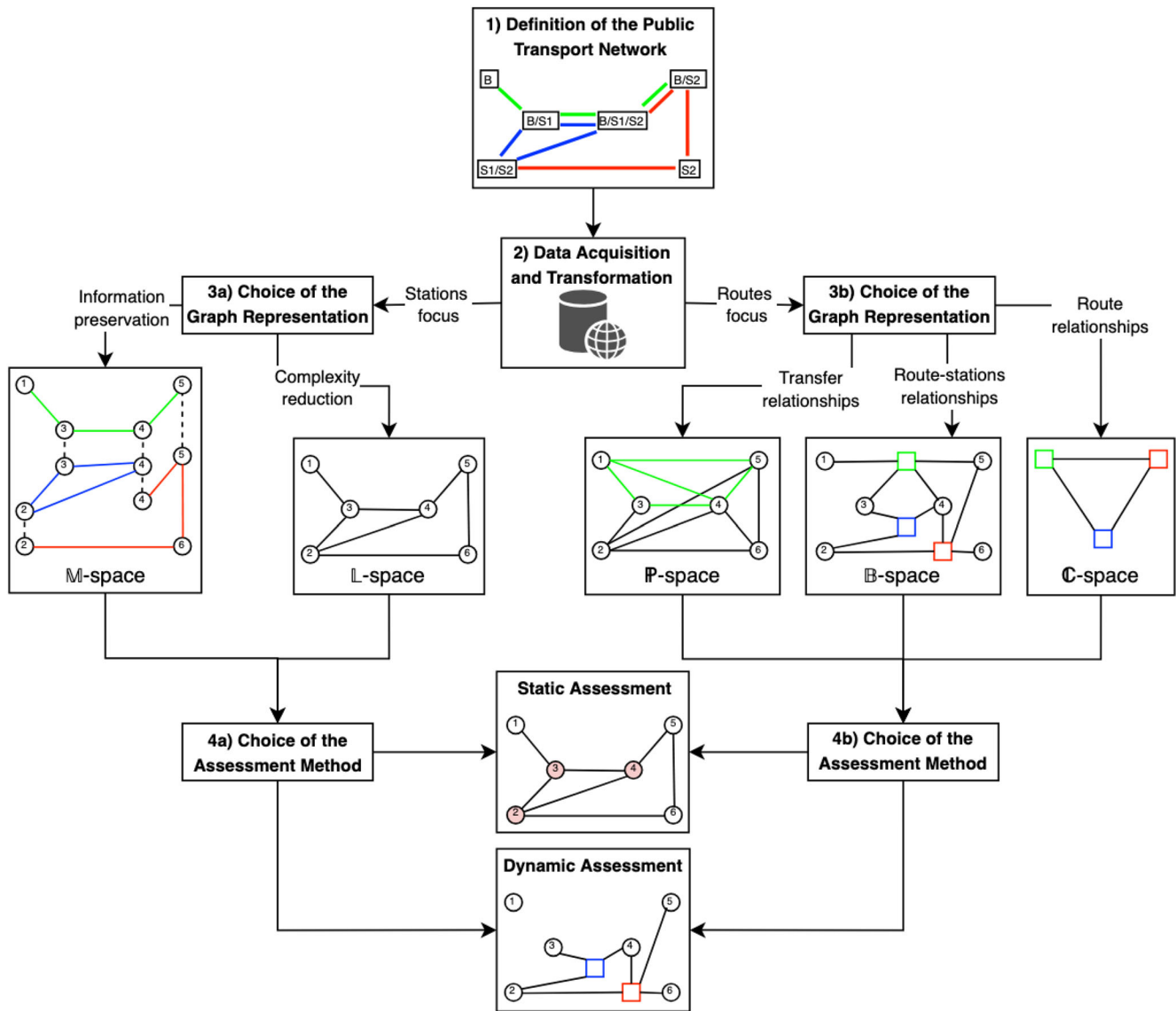


FIGURE 6 | Decision support for assessing robustness of PTNs in various graph representations.

single graph representation and static assessment, thus offering a more holistic perspective on PTNs.

- 1. Definition of the PTN:** The first step involves selecting the transportation modes and routes to be included in the PTN model. This decision determines the network's structure and scope for analysis. The motivation behind this choice could depend on which transportation modes or routes regularly show disturbances or are often affected by delays. Focusing on these problematic areas could be worth the deeper analysis, potentially revealing root causes and systemic bottlenecks. This targeted approach may yield valuable insights into which parts of the infrastructure should be improved to enhance the reliability and overall performance of the network.
- 2. Data acquisition and transformation:** Next, the availability of relevant data, such as GTFS datasets described in Section 3.1, has to be verified. Once obtained, the data is transformed into a format suitable for generating a graph

model. Hereby, the focus lies on GTFS due to its widespread, open availability and standardized format. Approaches relying on data from transportation agencies face significant limitations because such data is often not openly accessible or only available upon request, with no guarantee of approval. This restricts reproducibility and complicates applying these methods across different networks. In contrast, GTFS's open and standardized data format makes it possible to develop an approach that can be applied without restrictive data policies, enabling more consistent and transferable analysis across PTNs.

- 3. Choice of the graph representation:** Based on the defined PTN and the acquired data, an appropriate graph representation is selected. The choice depends on the objective of the robustness analysis. Based on the study's aim, (a) one or more of the $\mathbb{L}$ -, and $\mathbb{M}$ -space representations (see Section 3.2 and Section 3.3, respectively) for a focus on individual stations or (b) one or more of the $\mathbb{P}$ -, $\mathbb{B}$ -, and $\mathbb{C}$ -space

representations (see Section 3.4) for a focus on entire routes may be suitable.

4. **Choice of the assessment method:** Local and global graph properties are computed in the case of a static assessment to evaluate the robustness or to identify vulnerable nodes or edges in representations focusing (a) on stations or (b) on routes. By analyzing commonly used graph-based and spectral-based global indicators and comparing them with results from other PTNs in the literature, an overall robustness evaluation can be conducted. Furthermore, by comparing system robustness before and after construction, the impact of infrastructure investments can be evaluated. In Figure 6, a static assessment is exemplified for the PTN in $\mathbb{L}$ -space, where stations v_2 , v_3 , and v_4 are highlighted in red to indicate critical stations. To gain deeper insights into the robustness and vulnerability of a PTN, disturbance scenarios are simulated in a dynamic assessment by removing single nodes, edges, or subgraphs to mimic random or targeted disruptions. Also in this case, whether comparing two existing PTNs or evaluating planned PTN extensions, this approach can provide valuable insights into network resilience. The choice of graph representation plays a crucial role in shaping the outcomes and interpretations of these simulations. In Figure 6, a disturbance scenario is exemplified for the PTN in $\mathbb{B}$ -space, where the entire green route experiences a breakdown, resulting in missing connections between stations v_1 , v_3 , v_4 , and v_5 .

5 | Case Study for the Subway and Rapid Transit Railway System in Hamburg, Germany

The proposed decision-support approach is demonstrated through a case study on a real-world PTN, emphasizing robust assessment for long-term infrastructure planning using openly accessible data. This approach can be especially valuable because operational data, such as APC, are often difficult to obtain from transit agencies, highlighting the importance of assessing a PTN by means of usually available GTFS data. Such decision support for assessing PTN robustness can be of interest for city planners, as it enables the evaluation of PTNs by comparing various indicators with efficient designs of reference cities, using open data sources without being limited by the inaccessibility of operator-provided data. Furthermore, it can be used to evaluate the impact of infrastructure projects, such as line extensions, by comparing system robustness before and after construction.

5.1 | Definition of the Public Transport Network

In the first step, the PTN of interest and its associated transport modes are defined. For this case study, the subway and rapid transit railway network of Hamburg are selected, as illustrated in Figure 7 and Figure 8, respectively. The U-Bahn (subway) system, shown in the former figure, consists of four routes: U1, U2, U3, and U4. The S-Bahn (rapid transit railway) system, depicted in the latter figure, includes routes S1, S2, S3, and S5. The City of Hamburg plans to extend its U-Bahn system with a new line, the U5, scheduled to open by 2029, as illustrated in Figure 9. This

raises interest in understanding how such long-term infrastructure project affects the robustness of the overall subway and rapid transit network, including their ability to absorb disruptions and enhance the reliability of the PTN as a whole.

5.2 | Data Acquisition and Transformation

Since the HVV (Hamburger Verkehrsverbund) operator provides GTFS data,⁴ the network can be modeled based on this source as second step of the decision-support approach. In particular, the text files ‘routes.txt’, ‘trips.txt’, ‘stops.txt’, and ‘stop_times.txt’ are sufficient to construct a graph representation, as they detail the sequence of stations in a route. More specific, information about routes, their associated trips, the stops served, and the sequence in which stops are visited are extracted from these files.

First, all available routes are extracted and trips are filtered to include only those associated with these routes. Information about each stop is also loaded, including the stop name and geographic coordinates to enable visualization. Next, stop times are processed by grouping them according to their trip ID. For each trip, the corresponding sequence of stops is traversed and connections between consecutive stops are recorded as edges. All relevant stop IDs are collected during this process. Each edge is labeled with the name of the route it belongs to, and edges that are reused by the same route are counted accordingly. Once all stops and edges have been identified, they are added to the graph. Each stop is added as a node with associated metadata, and each connection between stops is added as an edge with attributes indicating the route and usage count. The use of a multi graph allows for multiple edges between the same pair of stops, which is essential for representing the PTN where different routes may follow overlapping paths.

5.3 | Choice of the Graph Representation

In the third step, the $\mathbb{L}$ - and $\mathbb{M}$ -space models are chosen for the graph representations to provide an aggregated view of the U-Bahn and S-Bahn system, offering insights into individual station connectivity and modal differences, as part of decision tree path 3(a). To compare the robustness of the U-Bahn and S-Bahn PTNs separately, the $\mathbb{L}$ -space representations of both networks are also analyzed. In addition, the effect of the U5 line extension is evaluated for the U-Bahn system specifically, as well as for the overall PTN. As part of decision path 3(b), the $\mathbb{B}$ - and $\mathbb{C}$ -space representations are applied to the U-Bahn system, focusing on the interplay between routes within the same mode. This multi-perspective approach enables robustness analysis at varying levels of granularity.

5.4 | Robustness Assessment of the U-Bahn and S-Bahn Network in $\mathbb{L}$ - and $\mathbb{M}$ -Space

Both the U-Bahn and S-Bahn networks are represented in $\mathbb{L}$ -space as separate PTNs. The aggregation of these networks results in a multimodal PTN in $\mathbb{L}$ -space representation. This representation is illustrated on the left-hand side in Figure 10. To analyze both modes as a multilayer graph, the $\mathbb{M}$ -space is utilized to represent



FIGURE 7 | U-Bahn subway network of Hamburg, consisting of the routes U1, U2, U3, and U4. Source: https://commons.wikimedia.org/wiki/File:Karte_der_U-Bahn_Hamburg.png (last accessed: 24.11.2025).

each network in separate layers. This representation is illustrated on the right-hand side in Figure 10.

5.4.1 | Static Assessment

To illustrate the step 4(a) of the decision-support approach, the robustness of the aggregated PTN, as well as the separated U-Bahn and S-Bahn modes in $\mathbb{L}$ -space, is evaluated by calculating and comparing robustness indicators for a static assessment. Furthermore, the U-Bahn PTN, including the planned U5 line extension, is considered. The results are presented in Table 7. In the following, we also compare the calculated metric values with the value ranges identified in the literature (see Table 4).

First, graph-based indicators are evaluated. The calculated values for the average degree, $\langle k \rangle \in [2.03, 2.18]$, and values for the degree diversity, $\kappa \in [2.16, 2.52]$, are similar to values reported in the literature ([1.93, 10.00] and [2.35, 7.91], respectively). However, κ is among the lower values, indicating less cohesive PTNs. The aggregated PTN yields slightly higher values for these metrics than the separated U-Bahn and S-Bahn PTNs, indicating increased robustness. Also, the planned U5 line extension to

the U-Bahn network positively impacts robustness, as both $\langle k \rangle$ and κ are higher compared to the current PTN. The scaling factors $\gamma \in [2.39, 2.52]$ are among the lower values compared to those from other studies ([2.10, 5.52]). Notably, $\gamma = 2.39$ for the aggregated PTN is slightly lower, indicating the presence of more high-degree stations. This aligns intuitively with the fact that central hubs accommodate routes from both modes in the aggregated representation. Degree assortativity values range from $[-0.0095, 0.1835]$, aligning with typical values in the literature. Interestingly, the clustering coefficient for the U-Bahn network is $\langle c \rangle = 0$, suggesting an absence of triangles in this network. In contrast, the planned extended network yields $\langle c \rangle = 0.0039$, indicating a more cohesive structure. The other calculated clustering values, in the range $[0.00, 0.0121]$, are comparable to those of other PTNs worldwide (being in the range of $[0.00, 0.161]$). The average shortest path length, $\langle l \rangle \in [11.87, 12.84]$, suggests relative efficient PTNs in comparison to the literature ([1.00, 53.50]). The U5 line extension also positively influences the U-Bahn network, as the reduced average path length of $\langle l \rangle = 11.24$ is lower than the current value of 11.87. With a network diameter of $\langle d \rangle \in [31, 37]$, the considered PTNs exhibit relatively low diameters, whereas the S-Bahn network has the lowest diameter of 31.



FIGURE 8 | S-Bahn rapid transit railway network of Hamburg, consisting of the routes S1, S2, S3, and S5. Source: https://commons.wikimedia.org/wiki/File:Karte_der_S-Bahn_Hamburg_2023.svg (last accessed: 24.11.2025).

For the spectral-based indicators, the natural connectivity, $\langle n \rangle \in [0.0063, 0.013]$, falls within the mid-range of reported values in the literature $[0.004, 0.22]$. The algebraic connectivity, $\langle \mu \rangle \in [0.000054, 0.000057]$, suggests relatively low robustness, as higher values indicate greater network resilience. The increased value of $\langle \mu \rangle = 0.00008$, compared to 0.000056, further suggests a potential improvement in robustness resulting from the planned U-Bahn line extension. In comparison, PTNs from other studies fall within $[0.00001, 0.073]$. The effective graph conductance, $C_G \in [0.00143, 0.00256]$, is also relatively low compared to the range of $[0.0009, 0.25]$ from other studies. Notably, the single U-Bahn and S-Bahn PTNs consistently exhibit higher values for all spectral-based indicators compared to the aggregated PTN.

The similarity of the calculated values suggests that it is not straightforward to determine which PTN is the most robust. Furthermore, according to certain indicators such as the average degree, the degree diversity, and the average shortest path length, the U-Bahn network appears to be more robust than the S-Bahn. Conversely, the S-Bahn exhibits higher values for spectral-based measures, indicating stronger robustness according to these metrics. These findings align with the discussion in [96], which

emphasizes the necessity of employing multiple methods and criteria when assessing PTN robustness. This need arises from the inherent difficulty in defining concepts like robustness and resilience, as well as the challenge of quantifying them into a single objective function for maximization. Therefore, the robustness of the PTNs is further evaluated through a dynamic assessment in the next step.

5.4.2 | Dynamic Assessment

After the static assessment, disturbances are simulated through various targeted attack scenarios, where nodes are sequentially removed, being the step 5(a) of the decision-support approach. Hereby, we compare the impact of such simulations on the aggregated PTNs in $\mathbb{L}$ - and $\mathbb{M}$ -space, as well as for the single U-Bahn, S-Bahn, and U-Bahn with planned U5 line extension in $\mathbb{L}$ -space. In five different scenarios, nodes with the highest (recalculated) degree, closeness, betweenness, and PageRank centrality, as well as randomly chosen nodes, are removed in each iteration. The impact on the PTN across different representations is assessed by tracking the size of the remaining LCC relative to the original size at each step. The results after 40 iterations are shown in Figure 11,

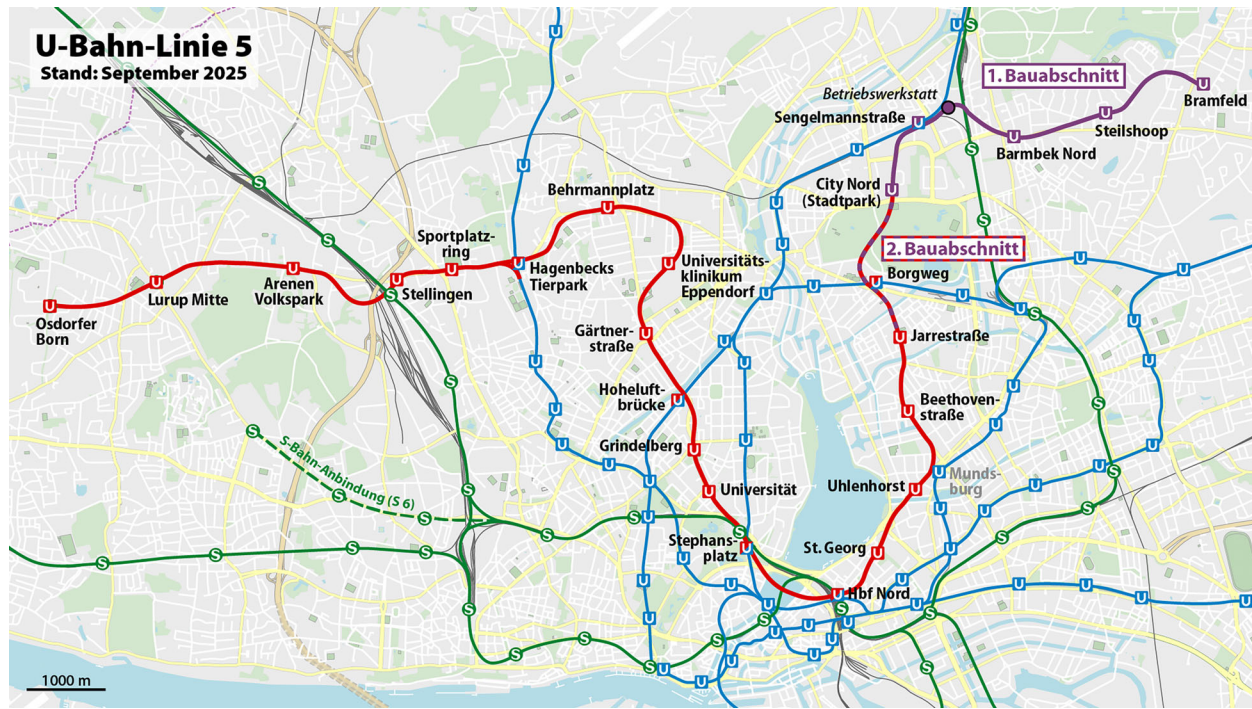


FIGURE 9 | Planned new U5 route for the U-Bahn subway network of Hamburg. Source: https://upload.wikimedia.org/wikipedia/commons/b/bf/Karte_U5_Hamburg.png (last accessed: 24.11.2025).

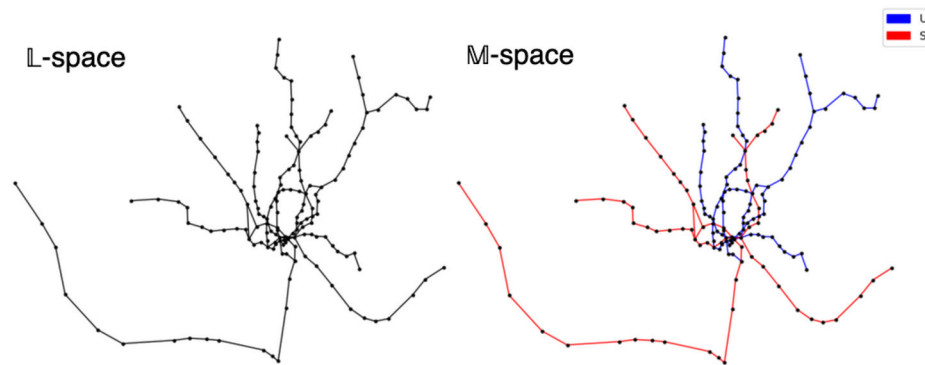


FIGURE 10 | Left: Aggregated U- and S-Bahn PTN of Hamburg in $\mathbb{L}$ -space. Right: Separate U- and S-Bahn network as layers of a multilayer graph in $\mathbb{M}$ -space, both based on geographical coordinates.

TABLE 7 | Comparison between global graph indicators of the aggregated PTN in $\mathbb{L}$ -space, as well as the single U-Bahn and S-Bahn PTNs in $\mathbb{L}$ -space.

Category	Indicator	$\mathbb{L}$ -space U-S	$\mathbb{L}$ -space U	$\mathbb{L}$ -space S	$\mathbb{L}$ -space U5
Graph-based	$ V $	154	93	69	110
	$ E $	168	98	70	119
	$\langle k \rangle$	2.18	2.11	2.03	2.16
	κ	2.52	2.30	2.16	2.40
	γ	2.39	2.55	2.41	2.41
	ρ	0.0074	-0.0095	0.1835	0.14
	$\langle c \rangle$	0.0054	0	0.0121	0.0039
Spectral-based	$\langle l \rangle$	12.48	11.87	12.08	11.24
	$\langle d \rangle$	37	33	31	33
	$\langle n \rangle$	0.0063	0.010	0.013	0.009
	$\langle \mu \rangle$	0.000054	0.000056	0.000057	0.00008
	C_G	0.00143	0.00235	0.00256	0.00235

Note: In addition, the U-Bahn PTN, including the planned U5 line extension, is considered.

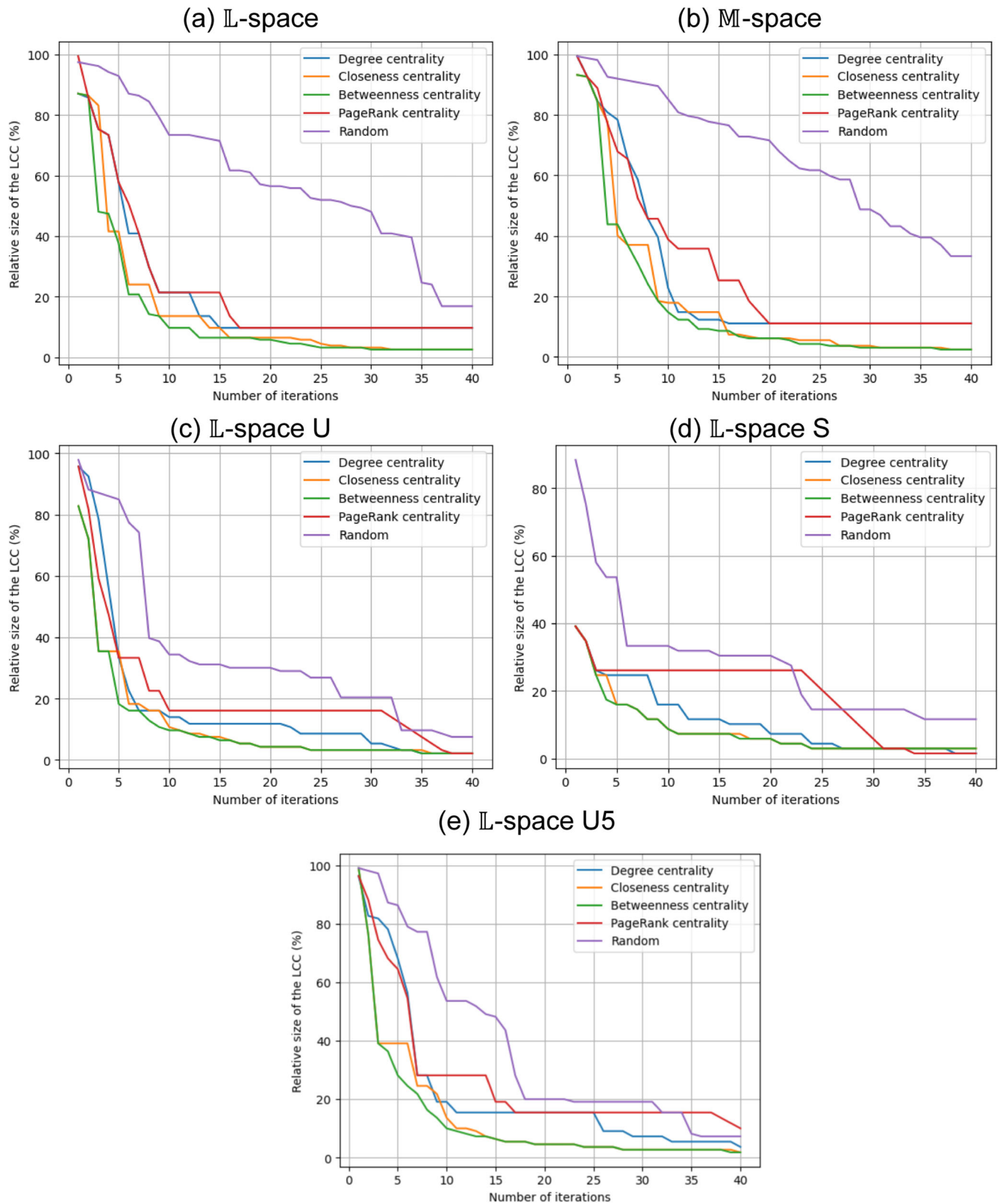


FIGURE 11 | Results of the simulation study for (a) aggregated PTN in L-space, (b) aggregated PTN in M-space, (c) single U-Bahn PTN in L-space, (d) single S-Bahn PTN in L-space, and (e) single U-Bahn PTN with planned U5 line extension, evaluated on the relative LCC size.

TABLE 8 | Sizes of the LCC relative to the original size (in percentages) across different representations after 3, 5, and 10 iterations (it.) of attack strategies based on various indicators.

No it	Attack strategy	Rel. LCC ($\mathbb{L}$ -space U-S)	Rel. LCC ($\mathbb{M}$ -space)	Rel. LCC ($\mathbb{L}$ -space U)	Rel. LCC ($\mathbb{L}$ -space S)	Rel. LCC ($\mathbb{L}$ -space U5)
3	Degree	75.32%	84.57%	78.49%	26.09%	81.82%
	Closeness	83.12%	84.57%	35.48%	24.64%	39.09%
	Betweenness	48.05%	84.57%	35.48%	24.64%	39.09%
	PageRank	75.32%	88.89%	59.14%	26.09%	74.55%
	Random	96.10%	98.15%	87.10%	57.97%	97.27%
5	Degree	57.79%	78.40%	33.33%	24.64%	68.18%
	Closeness	41.56%	40.12%	35.48%	15.94%	39.09%
	Betweenness	37.66%	43.83%	18.28%	15.94%	28.18%
	PageRank	57.79%	67.90%	33.33%	26.09%	64.55%
	Random	92.86%	91.98%	84.95%	53.62%	86.36%
10	Degree	21.43%	22.84%	13.98%	15.94%	19.09%
	Closeness	13.64%	17.90%	10.75%	8.70%	13.64%
	Betweenness	9.74%	14.81%	9.68%	8.70%	10.00%
	PageRank	21.43%	38.89%	16.13%	26.09%	28.18%
	Random	73.38%	85.19%	34.41%	33.33%	53.64%

Note: Bold values indicate the relative smallest sizes per iteration.

as well as after 3, 5, and 10 iterations in Table 8. In general, random node removal causes less disruption across all representations compared to the targeted removal of specific nodes, leading to a slower decline in connectivity. This observation aligns with the findings from all reviewed studies in this article. Furthermore, consistent with previous research [68, 77, 102, 122, 124, 131–133, 153] (in both $\mathbb{L}$ - and $\mathbb{M}$ -space), attacks targeting nodes with high betweenness centrality consistently result in the most rapid disintegration of the PTNs across all representations. This is intuitive for the considered PTNs, as transfer stations that connect multiple U- and S-Bahn routes are vulnerable points in the network.

In contrast to studies such as [20, 68, 70, 96–98, 112, 113, 126, 130, 150, 152] (in both $\mathbb{L}$ - and $\mathbb{M}$ -space), our findings suggest that targeting high-degree nodes causes less severe damage than focusing on nodes with high closeness centrality, which aligns with the results in [72]. The PageRank centrality, however, is the least effective indicator for identifying vulnerable parts of the PTN in this study, as attacks based on this metric result in a much slower decline in connectivity compared to other centrality measures. Among the reviewed literature, only [134] observed the most significant impact during disturbance simulations based on PageRank centrality.

When comparing the effects on the $\mathbb{L}$ - and $\mathbb{M}$ -space representations, we observe a similar behavior, as shown in Figure 4 and described in Table 5. The decline in the LCC size is slower in the $\mathbb{M}$ -space representation than in the $\mathbb{L}$ -space. Specifically, in the most disruptive simulations, where nodes with the highest betweenness centrality are targeted, the relative LCC size decreases as follows (see Table 8): After 3 iterations, it is 48.05% in $\mathbb{L}$ -space and 84.57% in $\mathbb{M}$ -space, after 5 iterations, 37.66% and 43.83%, and after 10 iterations, 9.74% and 14.81%, respectively. In

conclusion, the aggregated representation in $\mathbb{L}$ -space leads to an overestimation of disruption impacts.

Another insight which can be generated based on these results is that the single U-Bahn and S-Bahn PTNs are a lot less resilient against targeted attacks in comparison to the aggregated forms in $\mathbb{L}$ - and $\mathbb{M}$ -space. For instance, the relative LCC size is only 35.48% in the U-Bahn PTN and 24.64% in the S-Bahn PTN after 3 iterations for the attack strategy based on betweenness centrality. After 5 iterations, the size reduces already to 18.28% and 15.94%, respectively, which is less than half compared to the sizes in $\mathbb{L}$ - and $\mathbb{M}$ -space.

Regarding the impact of the U5 line extension on the U-Bahn PTN, the simulation results highlight the added value of this infrastructure measure in terms of network robustness. For instance, under the most severe attack strategies based on closeness and betweenness centrality, the relative size of the LCC is 39.09% with the U5 extension, compared to 35.48% without it after three iterations. Similarly, after five iterations, the LCC remains larger with the planned extension (28.18%) than without (18.28%) for attacks based on betweenness centrality. Also after ten iterations, the network with the U5 extension shows slightly improved robustness, with an LCC size of 10.00% compared to 9.68% without it.

5.5 | Robustness Assessment of the U-Bahn Network in $\mathbb{B}$ - and $\mathbb{C}$ -Space

Next, the four routes of the U-Bahn network are analyzed as PTNs in both $\mathbb{B}$ -space and $\mathbb{C}$ -space to focus on the routes rather than individual stations, illustrating step 4(b) of the decision tree. The routes U1, U2, U3, and U4 are also visualized in Figure 12, where

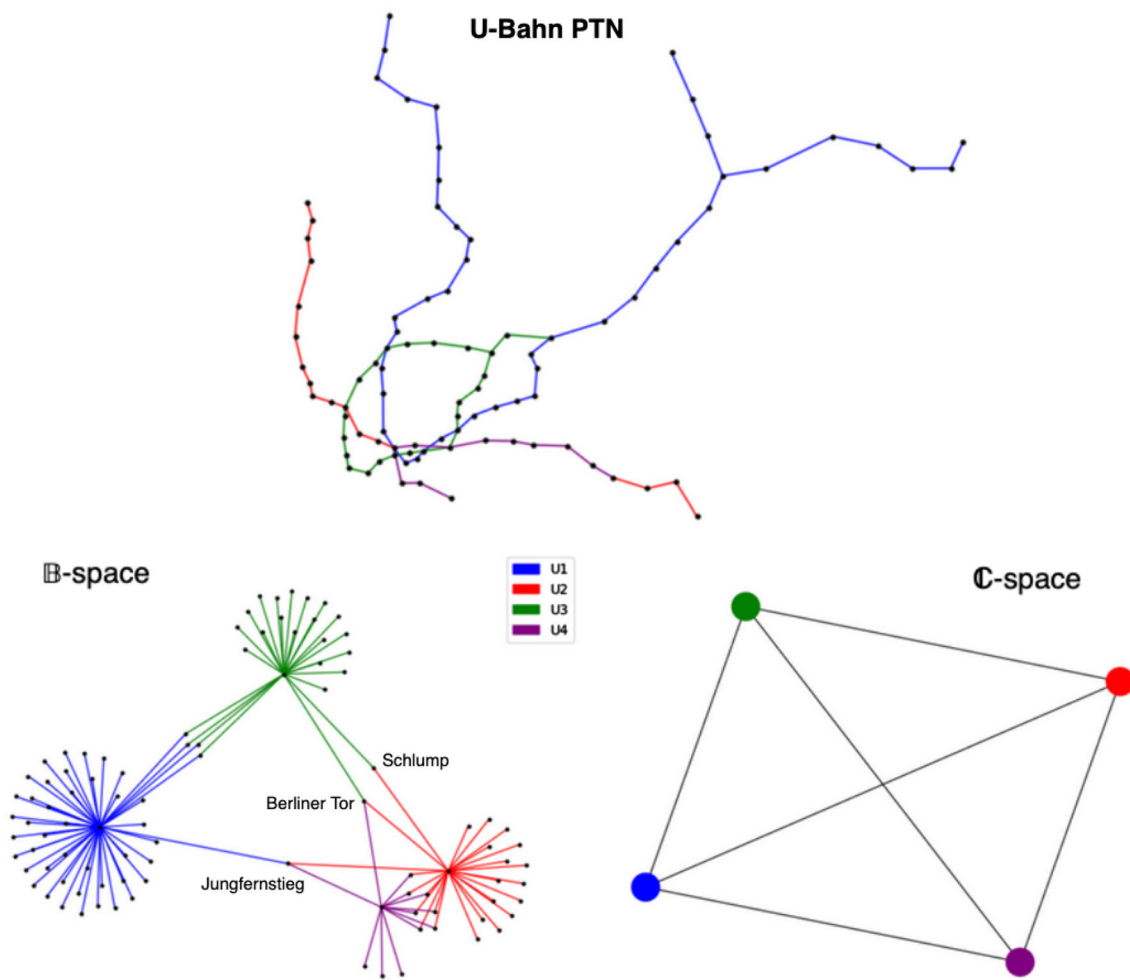


FIGURE 12 | U-Bahn PTN in $\mathbb{B}$ - and $\mathbb{C}$ -space representation, where the different routes U1, U2, U3, and U4 are shown in blue, red, green, and purple, respectively.

the routes in $\mathbb{B}$ -space and $\mathbb{C}$ -space can be more easily identified. Furthermore, to provide an outlook on how the future U5 route will impact the overall U-Bahn PTN, Figure 13 includes this line extension for infrastructure planning purposes.

In the most aggregated form of a PTN in $\mathbb{C}$ -space, it can be verified that each route is accessible through transfers at common stations, as seen in Figure 12. Since this representation results in a complete K^4 graph, the failure of an entire route does not affect the transfer possibilities between the remaining routes. Similarly to the complete K^5 graph in Figure 13, the new U5 line extension does not affect the original network's transferability.

The $\mathbb{B}$ -space representation in Figure 12 provides a finer level of granularity for the current U-Bahn network, allowing for the identification of important transfer stations. For example, “Jungfernstieg” is the only station connecting the routes U1, U2, and U4, such that a failure at this station would force passengers to take detours if they need to reach destinations on these routes. Another key hub is “Berliner Tor”, which connects U2, U3, and U4. The impact of a failure here may be less severe, as the routes U2 and U3 are still connected through “Schlump”. The vulnerability of such nodes with high betweenness centrality was also confirmed in the dynamic assessment. Transfers between U1

and U2, as well as U1 and U4, or U2 and U3,⁵ are only possible from one or two stations. In contrast, the connections between U2 and U4, as well as between U1 and U3 are more interconnected, meaning that a failure at a single station may be less disruptive for passengers due to the availability of alternative transfer options. These insights could not have been gained from the coarser $\mathbb{C}$ -space representation. Similarly, Figure 13 reveals new transfer stations between the current U-Bahn network and the planned U5 line extension. For example, “Hoheluftbrücke” would connect the U3 with the U5, “Hagenbecks Tierpark” would link the U2 with the U5, and “Sengelsmannstraße” would serve as an interchange between the U1 and the U5. Given the introduction of such new transfer stations through the planned U5 extension, policymakers should recognize that these stations are likely to become critical interconnection points within the U-Bahn network. As such, they may also represent potential points of vulnerability in the event of service disruptions, technical failures, or emergencies. Long-term infrastructure planning should, therefore, include robust contingency measures, such as redundancy in routing options, rapid-response protocols, and clear passenger communication strategies. Prioritizing resilience at these transfer hubs through enhanced monitoring, emergency evacuation procedures, and flexible operational designs could be essential to

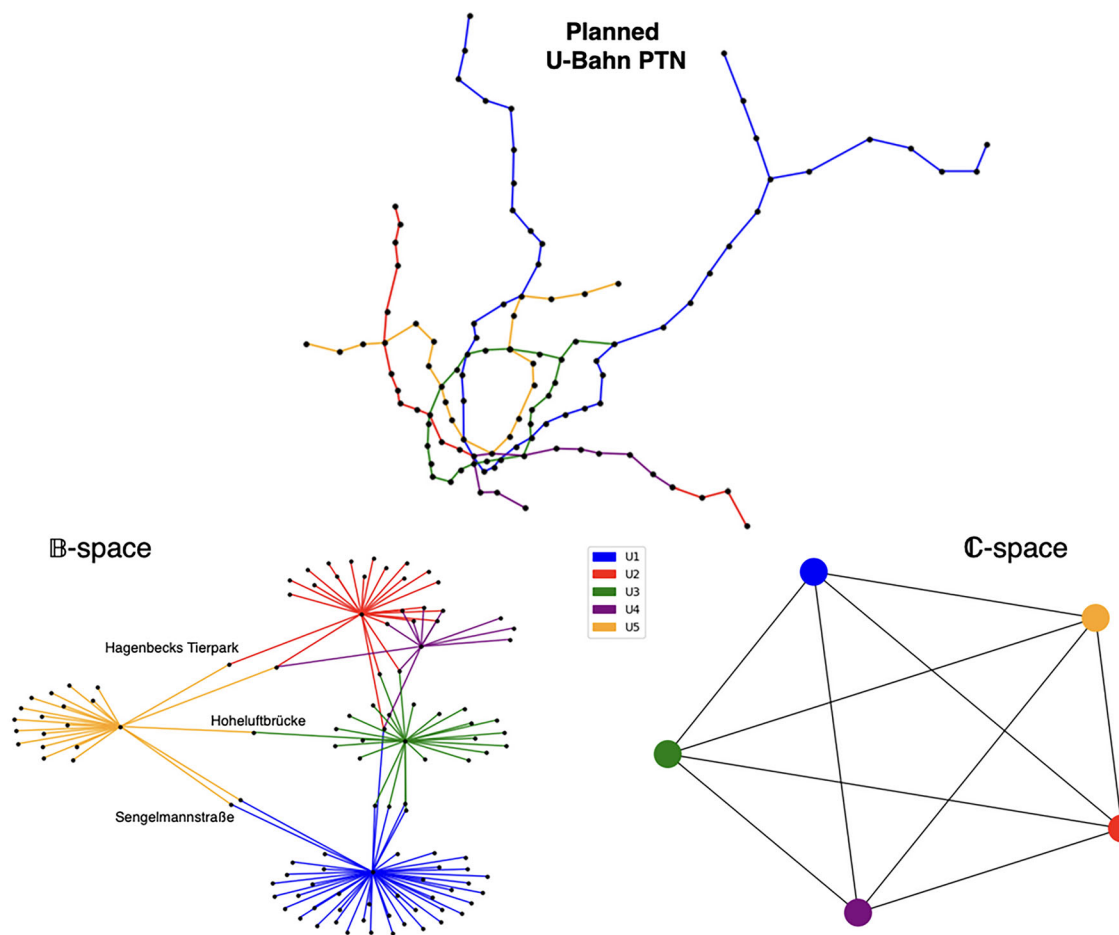


FIGURE 13 | Planned U-Bahn PTN in $\mathbb{B}$ - and $\mathbb{C}$ -space representation, where the different routes U1, U2, U3, U4, and U5 are shown in blue, red, green, purple, and orange, respectively.

ensuring uninterrupted network functionality and public safety in future scenarios.

6 | Conclusion

In this article, approaches for assessing the robustness of PTNs across various graph representations were reviewed. We find that different graph representations, namely $\mathbb{L}$ -, $\mathbb{M}$ -, $\mathbb{P}$ -, $\mathbb{B}$ -, and $\mathbb{C}$ -space, can yield different insights into the robustness of a PTN. The choice of the graph model depends on the objective of the robustness assessment, as well as the trade-off between complexity reduction and information preservation. While the $\mathbb{L}$ - and $\mathbb{M}$ -space representations focus on individual stations and connections within a PTN, the $\mathbb{P}$ -, $\mathbb{B}$ -, and $\mathbb{C}$ -space representations are more suited for analyzing entire routes. On one hand, the aggregation of nodes or edges in the $\mathbb{L}$ -space may fail to capture the multimodal characteristics of a PTN. On the other hand, the $\mathbb{M}$ -space representation preserves more of the PTN's information but increases model complexity. Similarly, the finer granularity of the $\mathbb{B}$ -space can reveal insights into the structure of a PTN that is not apparent in the coarser $\mathbb{C}$ -space graph model.

Furthermore, we proposed a decision-support approach for evaluating the robustness of a PTN in a holistic way. The decision

support begins with a clear definition of the PTN in question, followed by the acquisition of openly available data and the selection of appropriate graph representations based on the objectives of the robustness assessment. Hereby, the focus can be placed either on individual stations or on entire routes. Once the representations are chosen, the next step is to select the relevant graph indicators for evaluating the robustness in both static and dynamic assessments. In the static approach, various metrics are calculated to gain insights into the characteristics of the PTN, enabling comparisons between different networks. The dynamic approach, on the other hand, involves simulating disturbance scenarios across different graph representations to identify vulnerable parts of the PTN and assess the network's resilience during breakdowns, accidents, or delays. By integrating multiple assessment methods and graph representations, this approach provides a comprehensive way to evaluate the robustness of PTNs from various perspectives.

These concepts were applied in a case study of Hamburg's subway and rapid transit railway system, including an assessment of the potential impact of the planned U5 route extension on the U-Bahn network. For illustration purposes, the networks were represented in separated and aggregated form in $\mathbb{L}$ -, $\mathbb{M}$ -, $\mathbb{B}$ -, and $\mathbb{C}$ -space. Simulated disturbance scenarios revealed that the degradation process behaves differently depending on

the chosen representation. Additionally, the $\mathbb{B}$ -representation provides another perspective for assessing the different routes of the U-Bahn network, highlighting critical transfer stations that are essential for passenger movement between routes.

Future research could carefully compare different cities worldwide with respect to “critical infrastructure” within PTNs. Also the integration of additional transportation modes and dynamic factors such as passenger flow and temporal variations in the PTN could be explored across various graph representations, provided that operational data is made available by transportation agencies. Furthermore, analyzing real-world disruption scenarios alongside theoretical robustness measures would offer valuable insights and represents a promising research direction, especially if data on operational constraints, such as alternative route availability (see, e.g., [166]), and detailed passenger flow data (see, e.g., [12]) become publicly accessible in a standardized format. This would provide the opportunity to conduct similar analyses consistently across different cities, without limitations from restrictive transportation agencies’ data policies. Moreover, it would be interesting to check the network vulnerability regarding PTN users with restricted mobility and related usability of stations especially for transfer.

Author Contributions

Michael Palk: conceptualization; data curation; formal analysis; methodology; software; visualization; writing – original draft; writing – review and editing. **Stefan Voß:** conceptualization; supervision; writing – review and editing; validation. **Raka Jovanovic:** conceptualization; supervision; writing – review and editing; validation.

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Disclosure

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data are referenced in the manuscript.

Endnotes

¹<https://www.google.de/maps/> (last accessed: 24.11.2025).

²<https://www.openstreetmap.org> (last accessed: 24.11.2025).

³See, for example, <https://developers.google.com/transit> (last accessed: 24.11.2025) for technical documentation.

⁴<https://www.govdata.de/suche/daten/hvv-fahrplandaten-gtfs-januar-2025-bis-dezember-2025> (last accessed: 24.11.2025).

⁵Note that there is a direct underground connection making a short walk between the stations Rathaus (U3) and Jungfernstieg (U2, among others) possible even if they are indicated as two different stations.

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